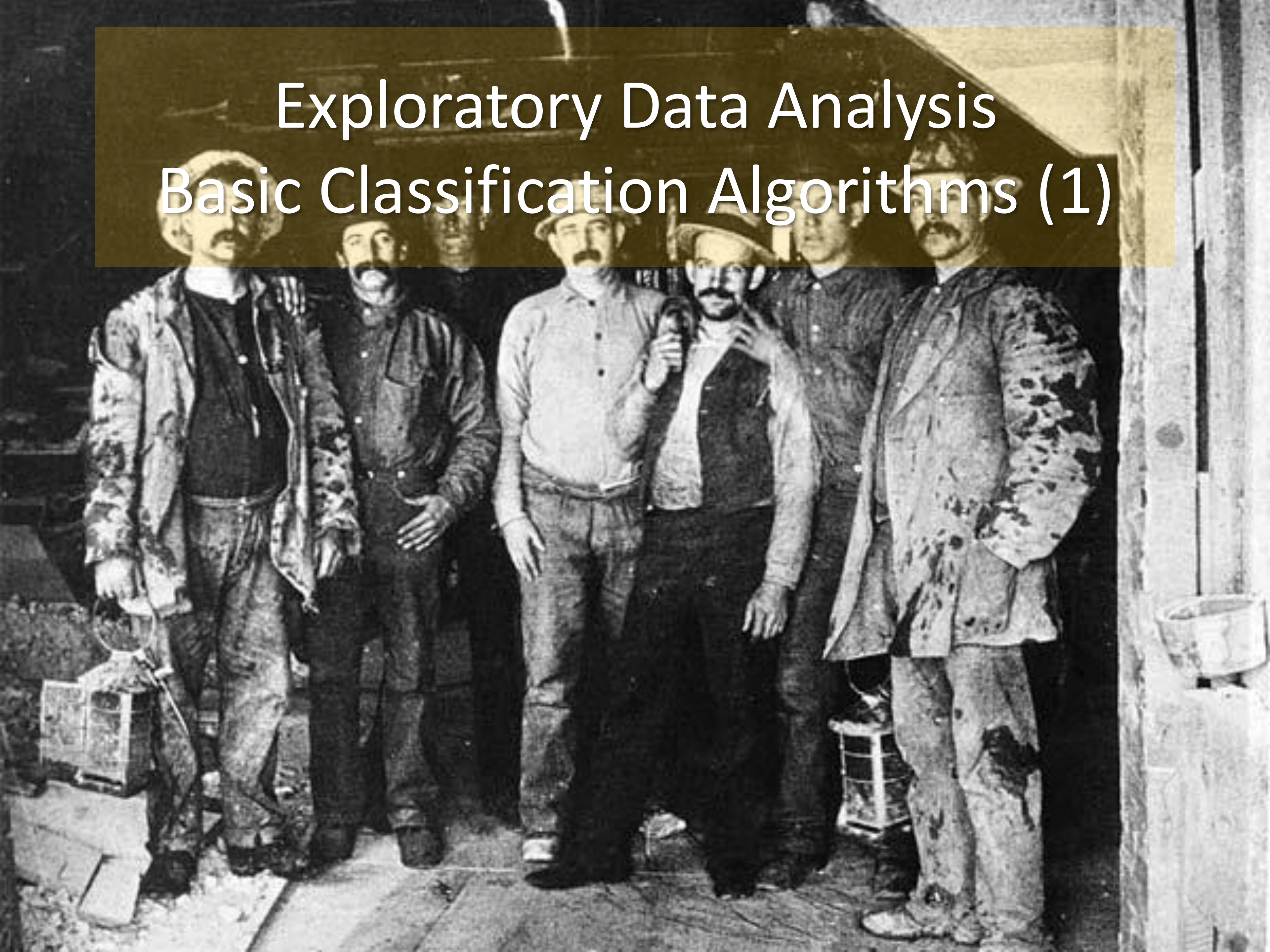
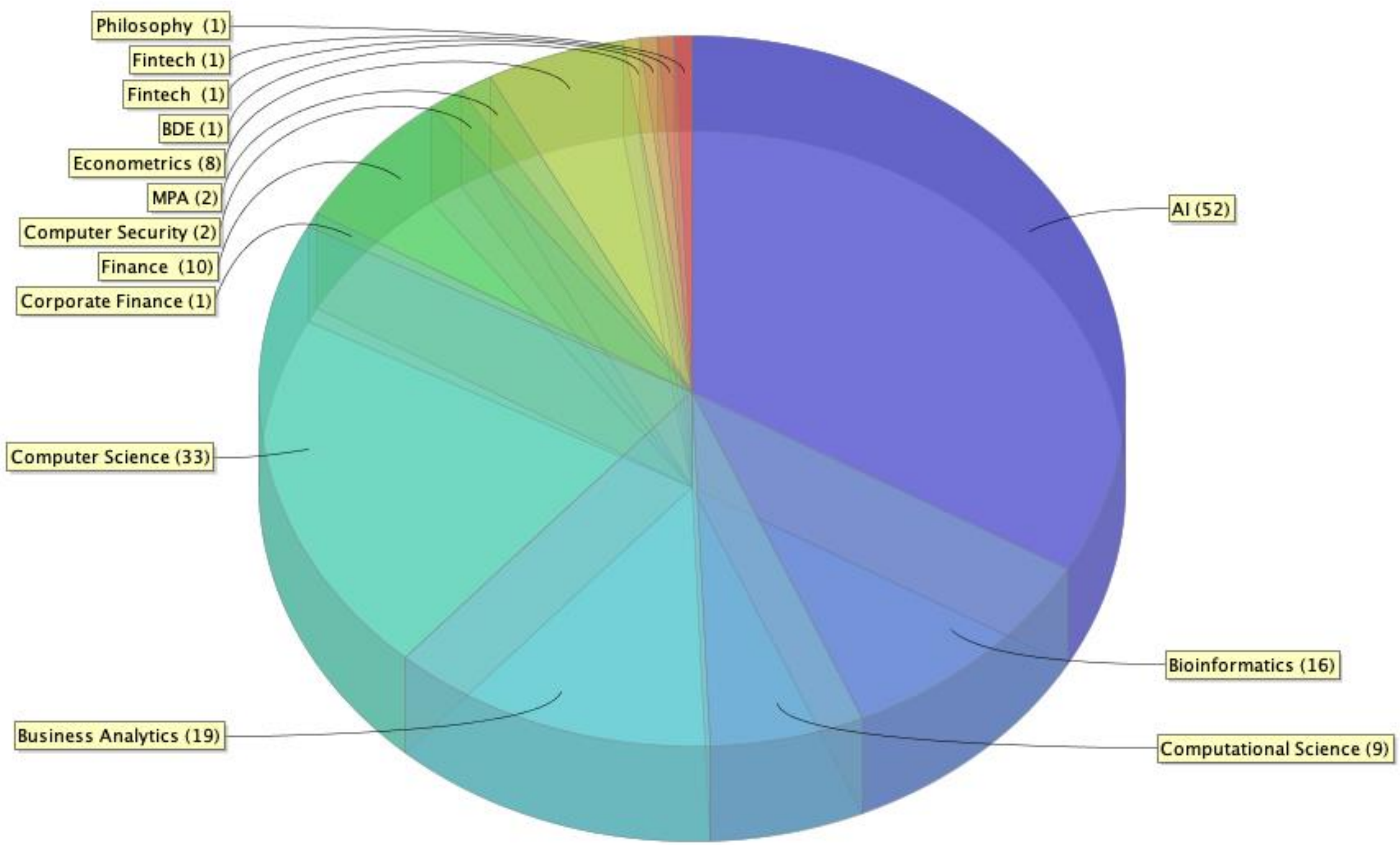
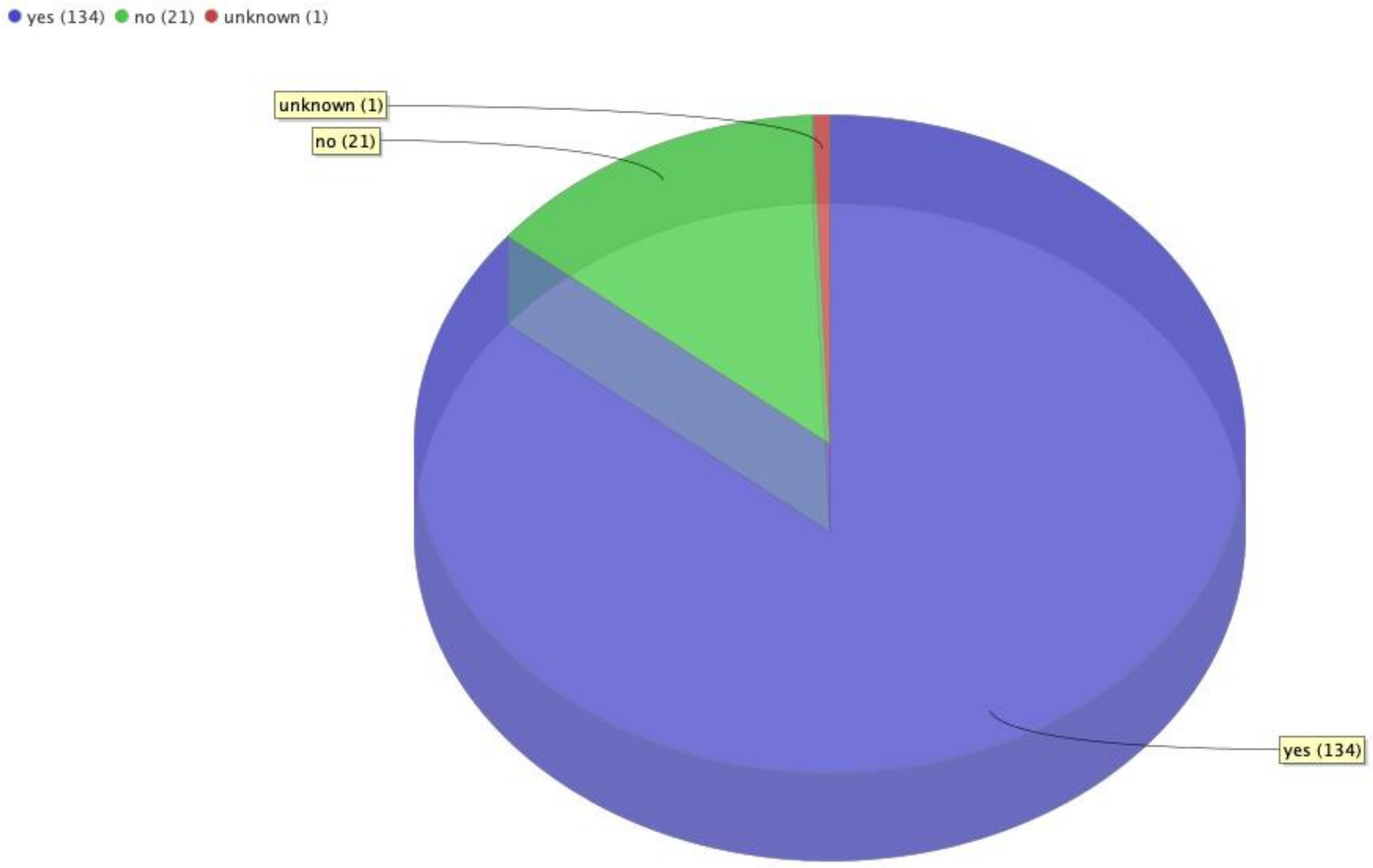
Exploratory Data Analysis
Basic Classification Algorithms (1)



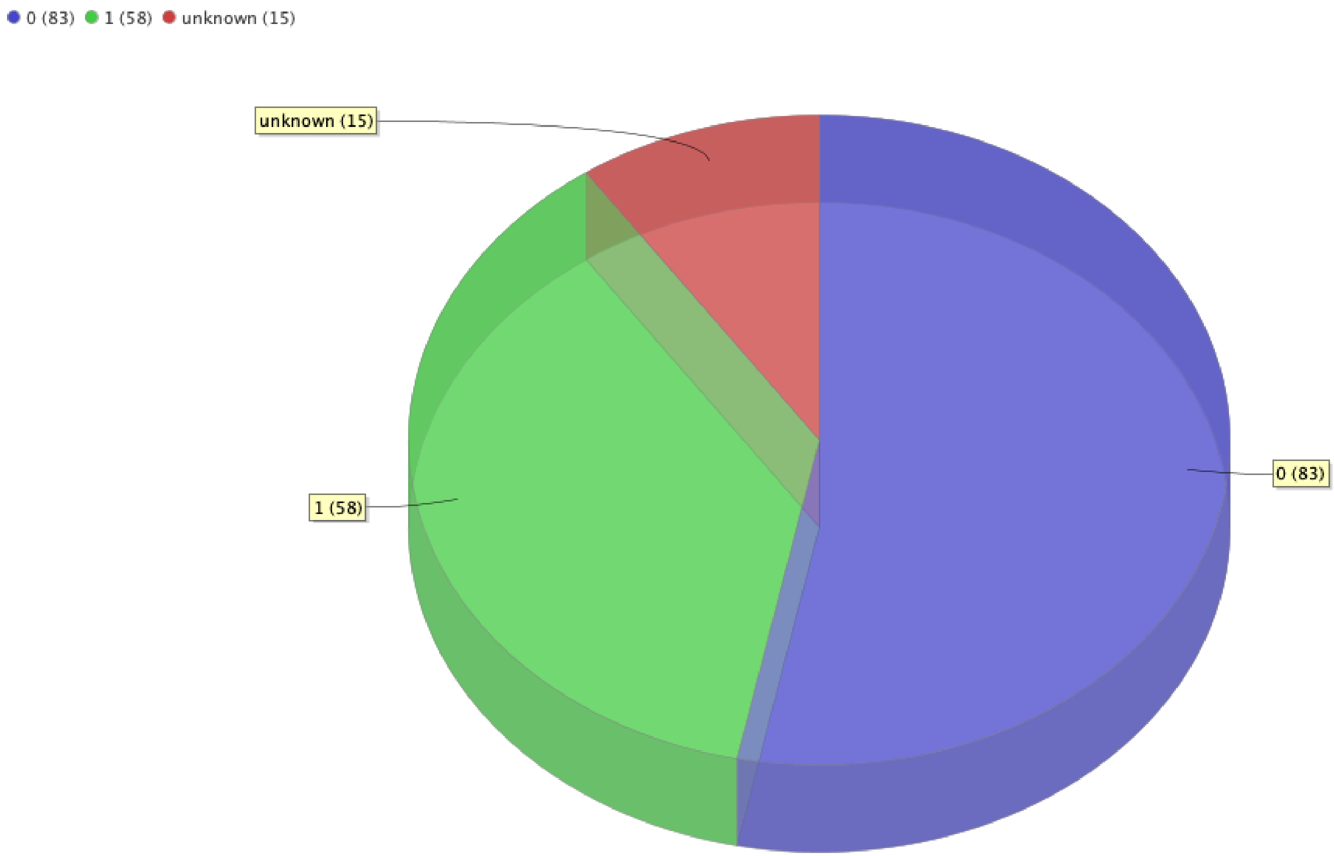
Your dataset



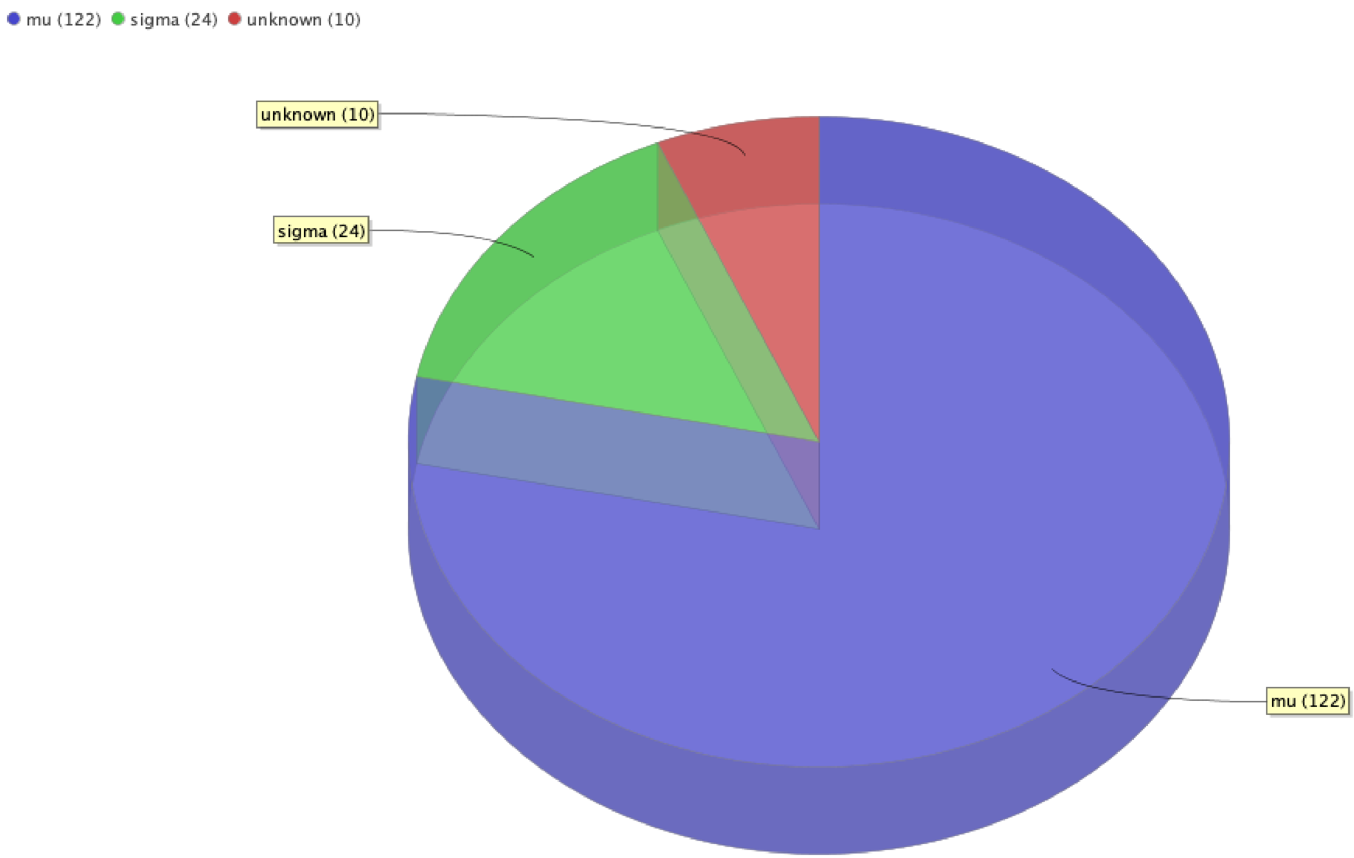
Machine Learning?



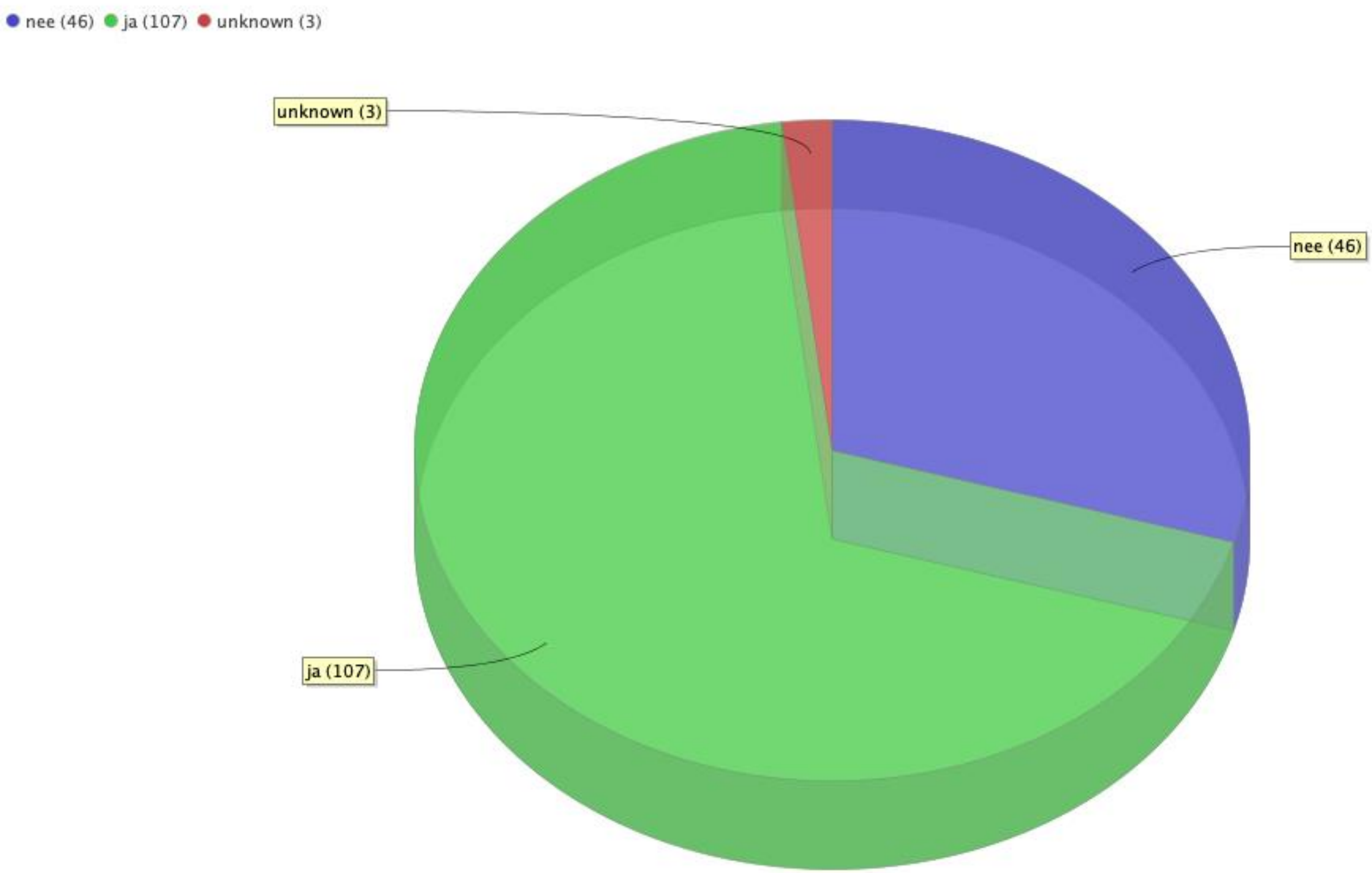
Information Retrieval?



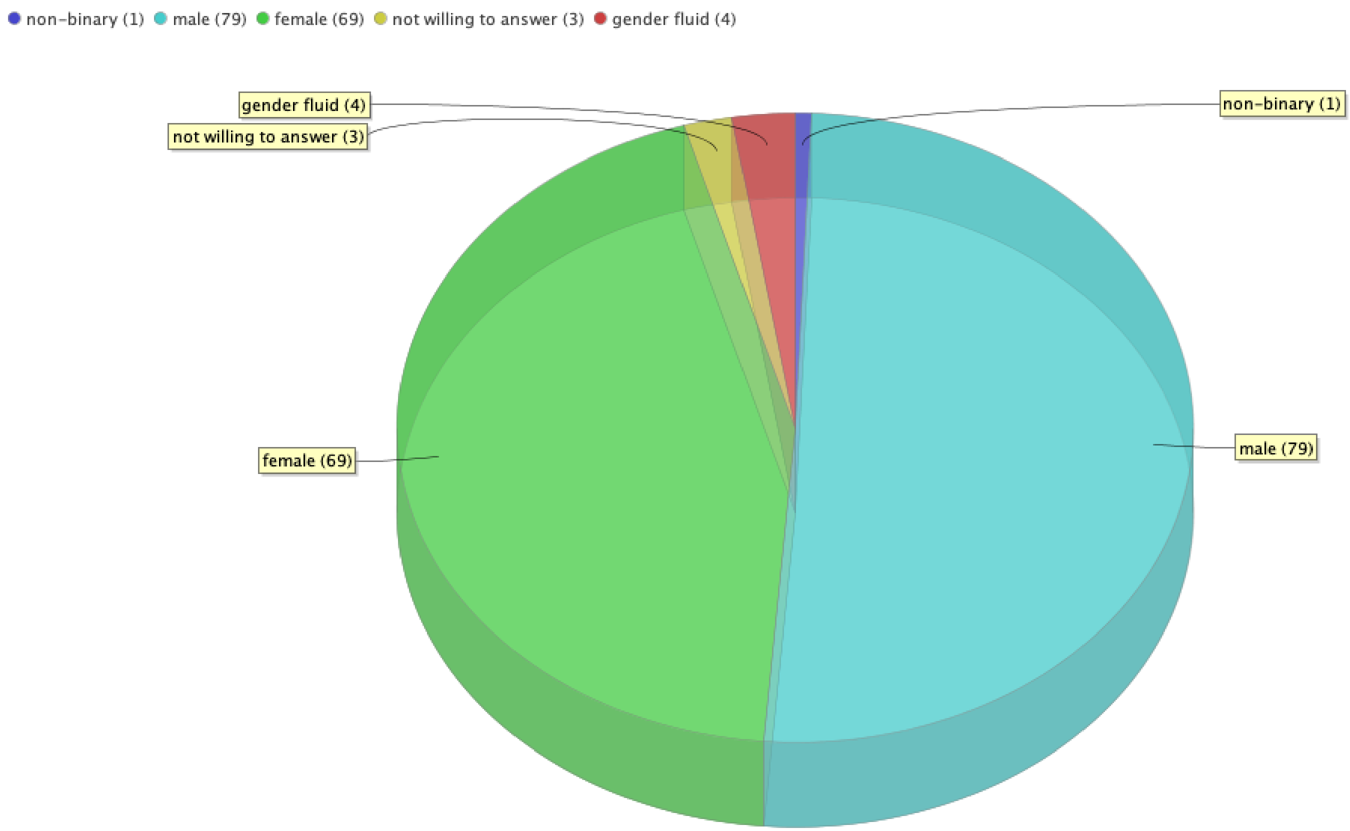
Statistics?



Databases?

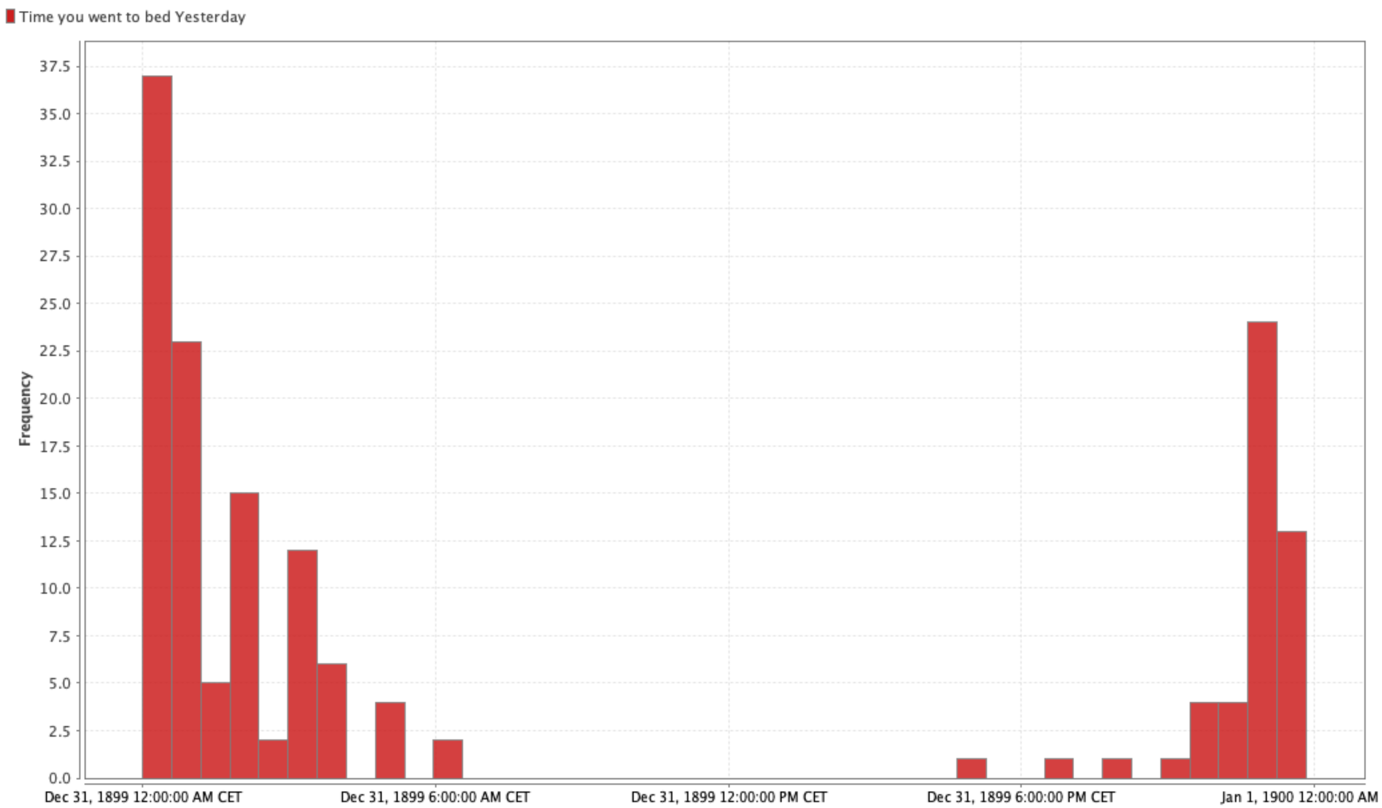


Gender?



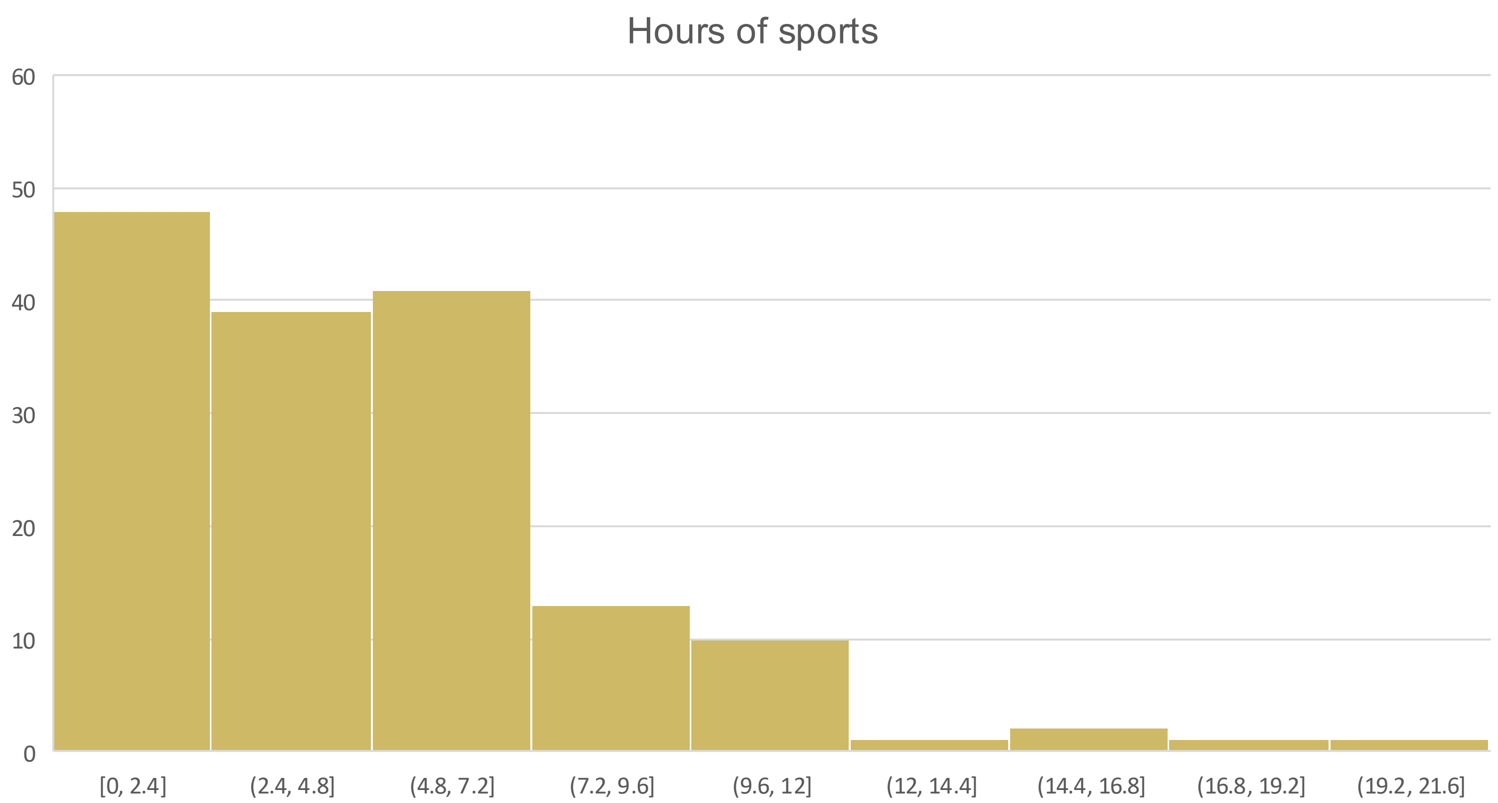
non-binary (1)
male (79)
female (69)
not willing to answer (3)
gender fluid (4)
gender fluid (4)
not willing to answer (3)
non-binary (1)
female (69)
male (79)

Bedtimes

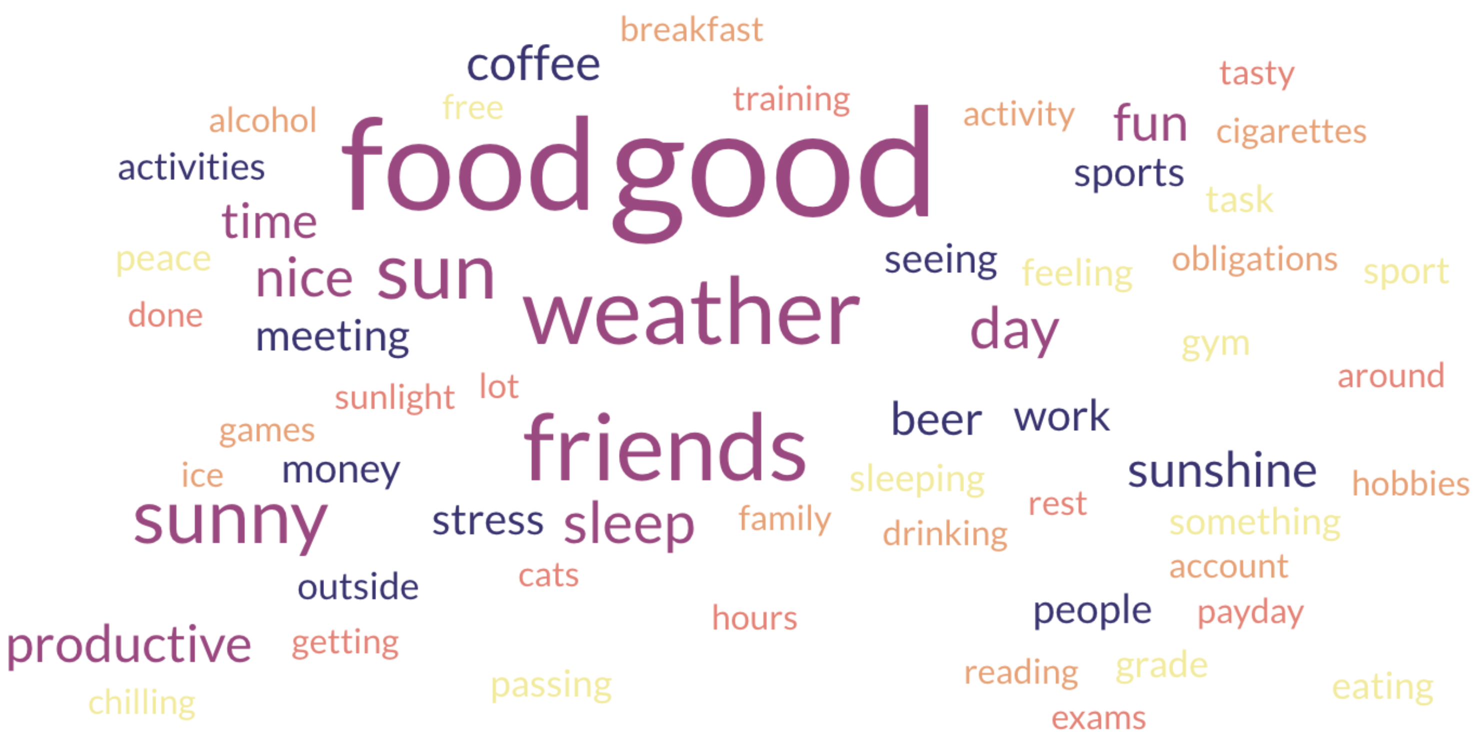


Time you went to bed Yesterday
Frequency
37.5
35.0
32.5
30.0
27.5
25.0
22.5
20.0
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0
Dec 31, 1899 12:00:00 AM CET
Dec 31, 1899 6:00:00 AM CET
Dec 31, 1899 12:00:00 PM CET
Dec 31, 1899 6:00:00 PM CET
Jan 1, 1900 12:00:00 AM

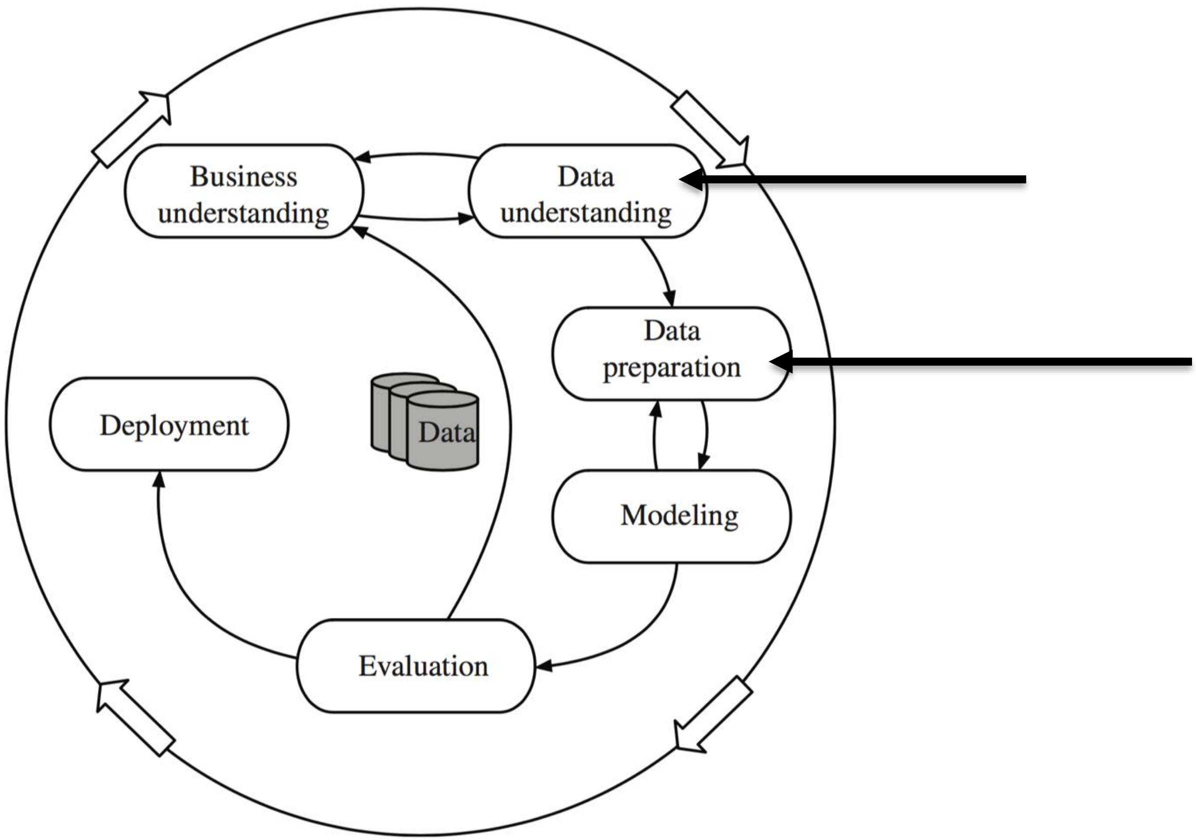
Sports


Hours of sports
60
50
40
30
20
10
0
[0, 2.4]
(2.4, 4.8]
(4.8, 7.2]
(7.2, 9.6]
(9.6, 12]
(12, 14.4]
(14.4, 16.8]
(16.8, 19.2]
(19.2, 21.6]

What makes a good day?



EDA


Business understanding
Data understanding
Data preparation
Deployment
Data
Modeling
Evaluation

EDA

- **EDA**: "detective work – numerical detective work – or counting detective work – or graphical detective work"

	A	B	C	D	E	F	G	H	I	J	K
1	File Name	EX_COEF_355	BA_COEF_355	EX_COEF_532	BA_COEF_532	BA_COEF_1064	mw	nm	real	imaginary	reff
2	FILE: r1200i000-0ln140410.LID d=0.01 m = 1.200 - .000E+00 i	2.02E-06	5.38E-09	1.56E-06	3.00E-09	2.70E-09	14	410	1.2	0.00E+00	5.44E-01
3	FILE: r1200i100-0ln140410.LID d=0.01 m = 1.200 - .100E+01 i	1.61E-06	9.46E-09	1.67E-06	9.56E-09	1.02E-08	14	410	1.2	1.00E+00	5.44E-01
4	FILE: r1200i100-1ln140410.LID d=0.01 m = 1.200 - .100E+00 i	1.57E-06	5.74E-10	1.41E-06	6.69E-10	9.98E-10	14	410	1.2	1.00E-01	5.44E-01
5	FILE: r1200i100-2ln140410.LID d=0.01 m = 1.200 - .100E-01 i	1.94E-06	2.56E-09	1.53E-06	2.06E-09	2.25E-09	14	410	1.2	1.00E-02	5.44E-01
6	FILE: r1200i100-3ln140410.LID d=0.01 m = 1.200 - .100E-02 i	2.01E-06	4.96E-09	1.56E-06	2.88E-09	2.65E-09	14	410	1.2	1.00E-03	5.44E-01
7	FILE: r1200i100-4ln140410.LID d=0.01 m = 1.200 - .100E-03 i	2.02E-06	5.34E-09	1.56E-06	2.99E-09	2.69E-09	14	410	1.2	1.00E-04	5.44E-01
8	FILE: r1200i100-5ln140410.LID d=0.01 m = 1.200 - .100E-04 i	2.02E-06	5.38E-09	1.56E-06	3.00E-09	2.70E-09	14	410	1.2	1.00E-05	5.44E-01
9	FILE: r1200i100-8ln140410.LID d=0.01 m = 1.200 - .100E-07 i	2.02E-06	5.38E-09	1.56E-06	3.00E-09	2.70E-09	14	410	1.2	1.00E-08	5.44E-01
10	FILE: r1200i125-1ln140410.LID d=0.01 m = 1.200 - .125E+00 i	1.53E-06	6.24E-10	1.40E-06	6.88E-10	9.65E-10	14	410	1.2	1.25E-01	5.44E-01
11	FILE: r1200i125-2ln140410.LID d=0.01 m = 1.200 - .125E-01 i	1.92E-06	2.18E-09	1.53E-06	1.89E-09	2.15E-09	14	410	1.2	1.25E-02	5.44E-01
12	FILE: r1200i150-1ln140410.LID d=0.01 m = 1.200 - .150E+00 i	1.51E-06	6.93E-10	1.40E-06	7.39E-10	9.79E-10	14	410	1.2	1.50E-01	5.44E-01
13	FILE: r1200i150-2ln140410.LID d=0.01 m = 1.200 - .150E-01 i	1.91E-06	1.88E-09	1.52E-06	1.75E-09	2.07E-09	14	410	1.2	1.50E-02	5.44E-01
14	FILE: r1200i175-1ln140410.LID d=0.01 m = 1.200 - .175E+00 i	1.49E-06	7.77E-10	1.40E-06	8.13E-10	1.03E-09	14	410	1.2	1.75E-01	5.44E-01
15	FILE: r1200i175-2ln140410.LID d=0.01 m = 1.200 - .175E-01 i	1.89E-06	1.63E-09	1.51E-06	1.63E-09	1.99E-09	14	410	1.2	1.75E-02	5.44E-01
16	FILE: r1200i200-1ln140410.LID d=0.01 m = 1.200 - .200E+00 i	1.48E-06	8.76E-10	1.41E-06	9.06E-10	1.10E-09	14	410	1.2	2.00E-01	5.44E-01
17	FILE: r1200i200-2ln140410.LID d=0.01 m = 1.200 - .200E-01 i	1.87E-06	1.43E-09	1.51E-06	1.52E-09	1.91E-09	14	410	1.2	2.00E-02	5.44E-01
18	FILE: r1200i225-1ln140410.LID d=0.01 m = 1.200 - .225E+00 i	1.47E-06	9.88E-10	1.42E-06	1.01E-09	1.20E-09	14	410	1.2	2.25E-01	5.44E-01
19	FILE: r1200i225-2ln140410.LID d=0.01 m = 1.200 - .225E-01 i	1.86E-06	1.27E-09	1.50E-06	1.42E-09	1.84E-09	14	410	1.2	2.25E-02	5.44E-01
20	FILE: r1200i250-1ln140410.LID d=0.01 m = 1.200 - .250E+00 i	1.47E-06	1.11E-09	1.42E-06	1.14E-09	1.31E-09	14	410	1.2	2.50E-01	5.44E-01
21	FILE: r1200i250-2ln140410.LID d=0.01 m = 1.200 - .250E-01 i	1.84E-06	1.14E-09	1.50E-06	1.33E-09	1.78E-09	14	410	1.2	2.50E-02	5.44E-01
22	FILE: r1200i250-3ln140410.LID d=0.01 m = 1.200 - .250E-02 i	2.00E-06	4.41E-09	1.55E-06	2.71E-09	2.57E-09	14	410	1.2	2.50E-03	5.44E-01
23	FILE: r1200i275-1ln140410.LID d=0.01 m = 1.200 - .275E+00 i	1.47E-06	1.25E-09	1.43E-06	1.27E-09	1.44E-09	14	410	1.2	2.75E-01	5.44E-01
24	FILE: r1200i275-2ln140410.LID d=0.01 m = 1.200 - .275E-01 i	1.83E-06	1.03E-09	1.49E-06	1.26E-09	1.72E-09	14	410	1.2	2.75E-02	5.44E-01

Goals of EDA

- Discover outliers ; missing values ; inconsistencies
- Data quality
- Get a ‘feeling’ of what is important and what is not
- Make first ‘discoveries’
- Get some feedback from domain experts
- Formulate first hypotheses

Steps of EDA

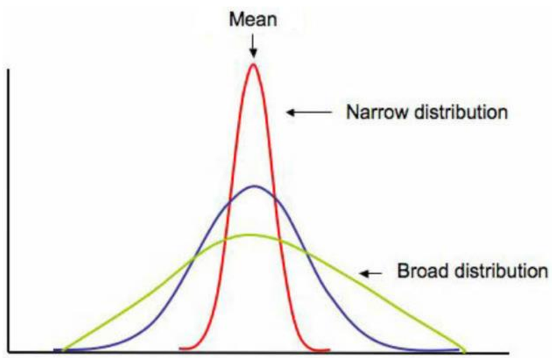
1. Data quality and logical structure
2. Data exploration
    a. Descriptive statistics
    b. Useful Plots
    c. Data Reduction
    d. Simple transformations
3. Importance & dependencies

1. Data quality & logical structure

- Structure:
  - "explicit" dependencies between variables (e.g., age vs. date_of_birth)
  - expected range of values (negative age?)
  - expected type of values (age as string?)
- Data quality
  - source of the data
  - is it up to date?
  - how was it collected and sampled?
  - "structural" inconsistencies
  - errors, outliers
  - missing values

2. Data exploration

- Count cases, #values, #missing values, ...
- Look at **location** and **spread** of the data
- Find distributions
- Make summaries
- Visualize the data
- Find dependencies
- Measure importance of attributes
- Data reduction
- Most important: ***think, think, think ...***



Steps of EDA

1. ~~Data quality and logical structure~~
2. Data exploration
   a. ~~Descriptive statistics~~
   b. ~~Useful Plots~~
   c. Data Reduction
   d. Simple transformations
3. Importance & dependencies

Data reduction

- Removal of data
- Transformations
- Aggregation of attributes
  - SUM / AVG
  - trends/coefficients/spectra...
  - principal components
  - standard features
- Missing values

Reduction options

- What can you reduce?
  - Number of rows (instances)
  - Number of columns (attributes)
- What can you use to reduce?
  - Rows, columns
    - Mostly focus on columns

2	4	5	7	7	2	4	5	7	7
2	3	34	9	2	2	3	34	9	2
4	4	5	4	7	4	4	5	4	7
5	5	3	2	6	5	5	3	2	6
4	2	4	5	7	4	2	4	5	7
2	26	56	7	2	2	26	56	7	2
2	6	7	2	5	2	6	7	2	5
26	6	7	2	7	26	6	7	2	7
2	4	5	7	7	2	4	5	7	7
2	3	34	9	2	2	3	34	9	2
4	4	5	4	7	4	4	5	4	7
5	5	3	2	6	5	5	3	2	6
4	2	4	5	7	4	2	4	5	7
2	26	56	7	2	2	26	56	7	2
2	6	7	2	5	2	6	7	2	5
26	6	7	2	7	26	6	7	2	7

Importance & dependencies

- It is always important to ***eliminate*** **insignificant** or **redundant** variables:
  - they introduce noise,
  - slow down algorithms,
  - decrease transparency of results,
  - lead to confusions, ...

- **Insignificant**: not contributing to final result
- **Redundant**: "equivalent" to another variable(s) (e.g., date of birth vs. age)

Are you sure it needs to go?

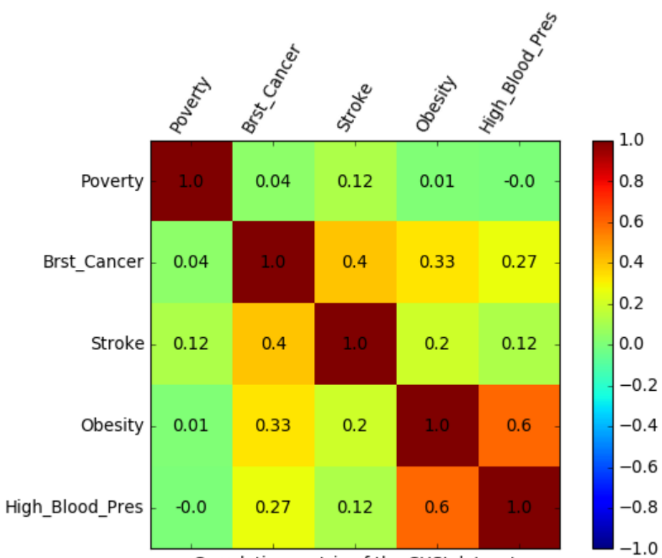
- In principle, an insignificant or redundant attribute must be removed.

HOWEVER

- A combination of unimportant attributes may be important

How to select what to remove

- **Common practice**:
  – identify a few "core attributes"



Correlation matrix of the CHSI dataset

Source: CHSI dataset

How to select what to remove

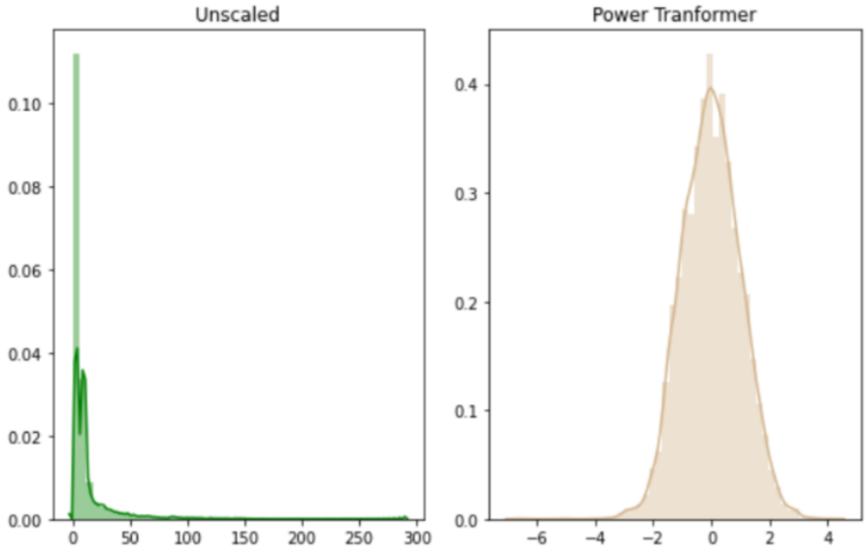
Top-down (backward)
Remove attributes one by one
For each candidate attribute “to be removed” 1. build a model on remaining attributes 2. remove the least influential attributes

How to select what to remove

Top-down (backward)	Bottom-up (forward)
Remove attributes one by one	Add attributes one by one
For each candidate attribute "to be removed" 1. build a model on remaining attributes 2. remove the least influential attributes	1. Start with individual attributes 2. Select the most influential one 3. Add all remaining attributes to the selected set 4. Keep the most influential one (and add to prev. selection)

Not always removal: Transformations

- **Scaling** (linear, exp, log, logsig)
  - Easier to see relationships in plots
  - Might work better when modelling

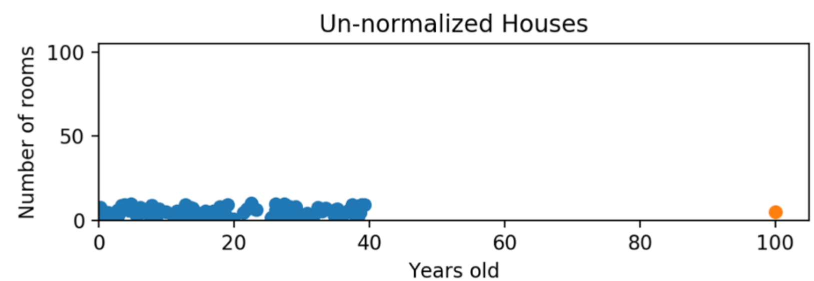


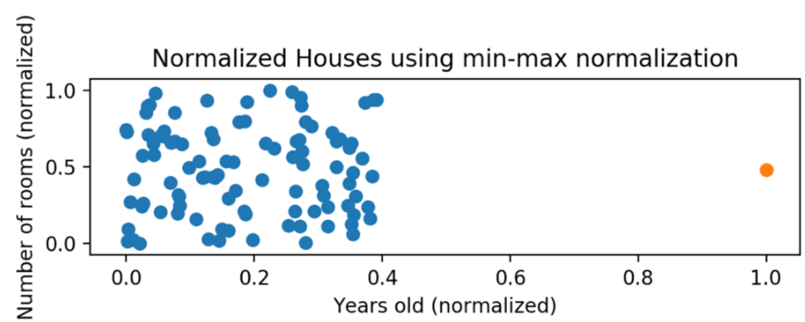
Not always removal: Transformations

- **Discretisation**
  - Only a couple of possible categories might remain
- **Recoding**
  - Long category strings to numbers *(**reduces** used memory!)*
- **Grouping** categorical values *(e.g., recoding postal codes into: big cities, small cities, villages)*
  - **Reduction** in number of categories!
- **Etc.**

Transformations

- Normalisation
  - Divide by maximum value *(max will be 1)*
  - Perhaps shift the values to [0,1]
- Standardisation
  - Centre data around zero (values - mean)
  - Divide by standard deviation





Aggregation of attributes

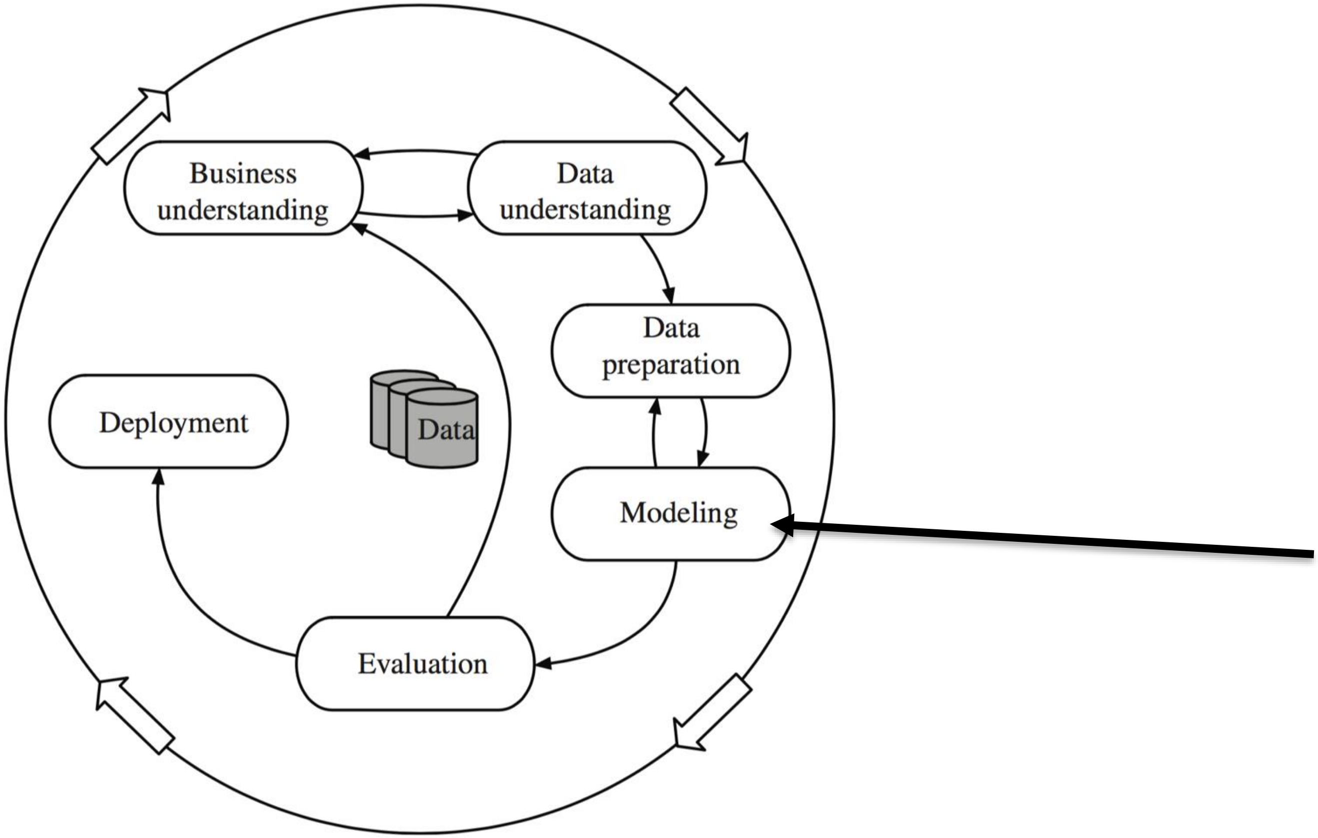
- Aggregation: combining several (meaningless) attributes into a meaningful one
- Examples of domain specific aggregates:
  - payments and withdrawals *(2 →1)*
  - average nr. of transactions over the last 6 months
  - “degree of satisfaction”: how well did I perform compared to others?

Domain-specific reduction/transformation

In many domains there are standard features that should be used as attributes:

- Text:
  - bi- and tri-grams
  - frequencies of “important-words”
- Sound:
  - power spectral density (PSD)
  - Fourier spectra
  - Envelopes
- Image:
  - visual features
  - various transforms

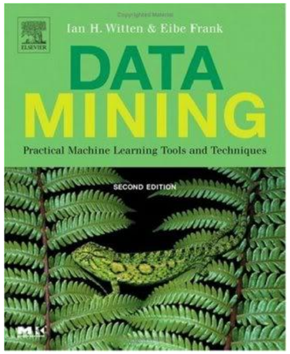
Basic DM algorithms for classification



Basic DM algorithms for classification

- 1R
- Naive Bayes
- Decision Trees
- Information Gain

Sections 4.1-4.4 of Witten & Frank



The weather data

Table 1.2 **The weather data.**

outlook	temperature	humidity	windy	play
sunny	hot	high	false	no
sunny	hot	high	true	no
overcast	hot	high	false	yes
rainy	mild	high	false	yes
rainy	cool	normal	false	yes
rainy	cool	normal	true	no
overcast	cool	normal	true	yes
sunny	mild	high	false	no
sunny	cool	normal	false	yes
rainy	mild	normal	false	yes
sunny	mild	normal	true	yes
overcast	mild	high	true	yes
overcast	hot	normal	false	yes
rainy	mild	high	true	no

The weather problem

- Input:

  Current weather conditions

  (outlook, temperature, humidity, wind)

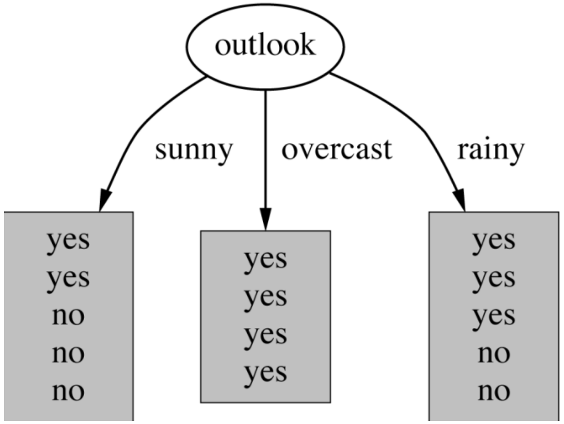
- Output:

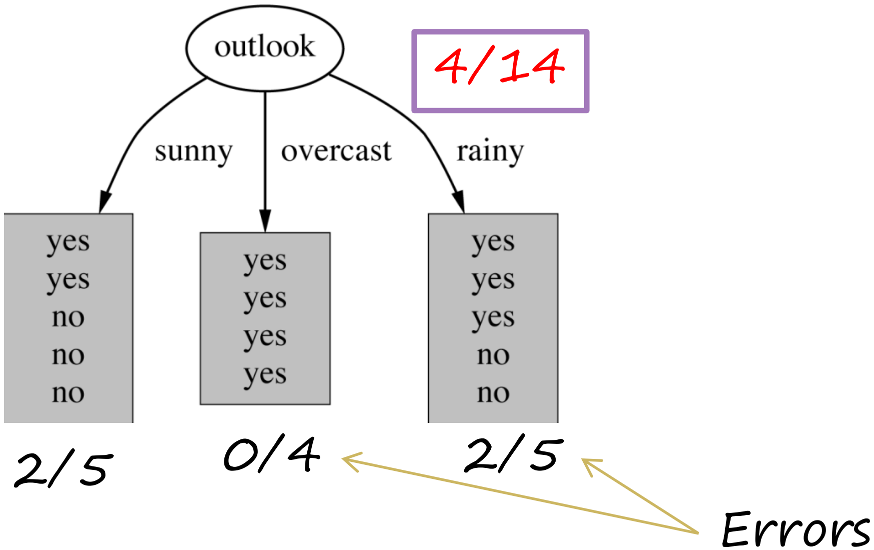
  To play or **not to play**?

- **Task:** Given some historical data, find some general **decision rules** about when to play

The One-rule (1R) algorithm

- Generates a one level decision tree
- Idea:
  - Take an attribute
  - Take all its possible values
  - For each value, find out which value of the output class is most frequent, and assign that class value to this attribute value.
  - Do this for all attributes
  - Choose the rules of the one making the least number of mistakes


outlook
sunny
overcast
rainy
yes
yes
no
no
no
yes
yes
yes
yes
yes
yes
yes
no
no


outlook
4/14
sunny
overcast
rainy
yes
yes
no
no
no
yes
yes
yes
yes
yes
yes
yes
no
no
2/5
0/4
2/5
Errors


outlook
sunny
overcast
rainy
yes yes no no no
yes yes yes yes
yes yes yes no no
2/5
0/4
2/5
4/14
temperature
hot
mild
cool
yes yes no no
yes yes yes yes no no
yes yes yes no
2/4
2/6
1/4
5/14
Errors
humidity
high
normal
yes yes yes no no no no
yes yes yes yes yes yes no
3/7
1/7
4/14
windy
false
true
yes yes yes yes yes yes no no
yes yes yes no no no
2/8
3/6
5/14
Total errors

Weather & 1R in a table

Table 4.1 **Evaluating the attributes in the weather data.**

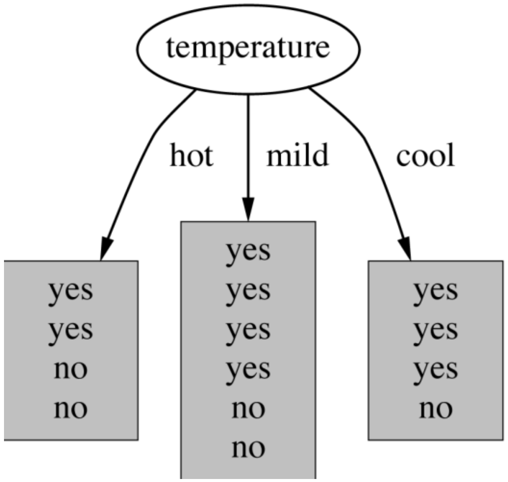
	attribute	rules	errors	total errors
1	outlook	sunny → no	3/5	4/14
		overcast → yes	0/4	
		rainy → yes	2/5	
2	temperature	hot → no*	2/4	5/14
		mild → yes	2/6	
		cool → yes	1/4	
3	humidity	high → no	3/7	4/14
		normal → yes	1/7	
4	windy	false → yes	2/8	5/14
		true → no*	3/6	

Weather data with numerical attributes

Table 1.3 Weather data with some numeric attributes.

outlook	temperature	humidity	windy	play
sunny	85	85	false	no
sunny	80	90	true	no
overcast	83	86	false	yes
rainy	70	96	false	yes
rainy	68	80	false	yes
rainy	65	70	true	no
overcast	64	65	true	yes
sunny	72	95	false	no
sunny	69	70	false	yes
rainy	75	80	false	yes
sunny	75	70	true	yes
overcast	72	90	true	yes
overcast	81	75	false	yes
rainy	71	91	true	no

Weather data with numerical attributes



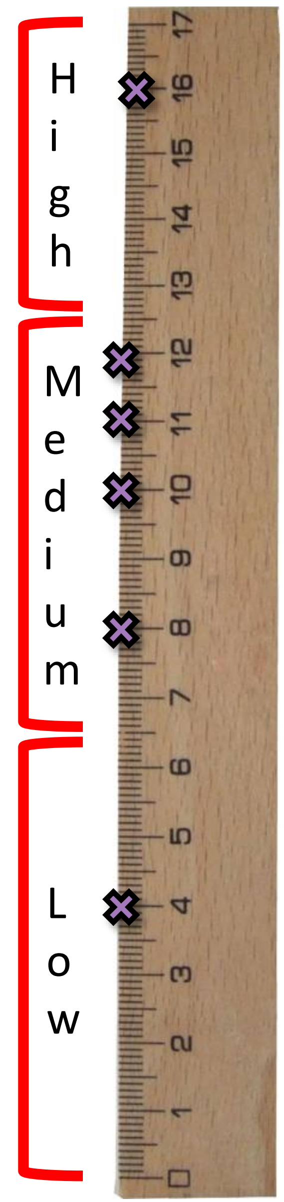
Too many specific cases.
Overfitting

Table 1.3 Weather data with some numeric attributes.

outlook	temperature	humidity	windy	play
sunny	85	85	false	no
sunny	80	90	true	no
overcast	83	86	false	yes
rainy	70	96	false	yes
rainy	68	80	false	yes
rainy	65	70	true	no
overcast	64	65	true	yes
sunny	72	95	false	no
sunny	69	70	false	yes
rainy	75	80	false	yes
sunny	75	70	true	yes
overcast	72	90	true	yes
overcast	81	75	false	yes
rainy	71	91	true	no

Numeric to Nominal

- Sometimes we need to turn numbers into categories
  - To avoid overfitting
  - To be able to deal with intermediate values
    - Temperature 15.123 (for example)
- The process is called discretisation



Discretising temperature in the weather data

Table 1.3 Weather data with some numeric attributes.

outlook	temperature	humidity	windy	play
sunny	85	85	false	no
sunny	80	90	true	no
overcast	83	86	false	yes
rainy	70	96	false	yes
rainy	68	80	false	yes
rainy	65	70	true	no
overcast	64	65	true	yes
sunny	72	95	false	no
sunny	69	70	false	yes
rainy	75	80	false	yes
sunny	75	70	true	yes
overcast	72	90	true	yes
overcast	81	75	false	yes
rainy	71	91	true	no

- Step 1: order attribute values and find accompanying class

64	65	68	69	70	71	72	72	75	75	80	81	83	85
yes	no	yes	yes	yes	no	no	yes	yes	yes	no	yes	yes	no

Discretising temperature in the weather data

- Step 1: order attribute values and find accompanying class
- Step 2: Find breaking points in this sequence:
  - Per class change

64	65	68	69	70	71	72	72	75	75	80	81	83	85
yes	no	yes	yes	yes	no	no	yes	yes	yes	no	yes	yes	no

Discretising temperature in the weather data

- Step 1: order attribute values and find accompanying class
- Step 2: Find breaking points in this sequence:
  - Per class change

64	65	68	69	70	71	72	72	75	75	80	81	83	85
yes	no	yes	yes	yes	no	no	yes	yes	yes	no	yes	yes	no

Discretising temperature in the weather data

- Step 1: order attribute values and find accompanying class
- Step 2: Find breaking points in this sequence:
  - Per class change

64	65	68	69	70	71	72	72	75	75	80	81	83	85
yes	no	yes	yes	yes	no	no	yes	yes	yes	no	yes	yes	no

Discretising temperature in the weather data

- Step 1: order attribute values and find accompanying class
- Step 2: Find breaking points in this sequence:
  - Per class change

64	65	68	69	70	71	72	72	75	75	80	81	83	85
yes	no	yes	yes	yes	no	no	yes	yes	yes	no	yes	yes	no

Discretising temperature in the weather data

- Step 1: order attribute values and find accompanying class
- Step 2: Find breaking points in this sequence:
  - Per class change (good idea?)

64	65	68	69	70	71	72	72	75	75	80	81	83	85
yes	no	yes	yes	yes	no	no	yes	yes	yes	no	yes	yes	no

Discretising temperature in the weather data

- Step 1: order attribute values and find accompanying class
- Step 2: Find breaking points in this sequence:
  - Per class change (good idea?)

64	65	68	69	70	71	72	72	75	75	80	81	83	85
yes	no	yes	yes	yes	no	no	yes	yes	yes	no	yes	yes	no

Not taking
‘neighborhood’
into account

Discretising temperature in the weather data

- Step 1: order attribute values and find accompanying class
- Step 2: Find breaking points in this sequence:
  - Per class change
  - Impose a minimum number of cases for the majority class
    - O(3)
    - When adjacent partitions have the same majority class they can be merged.

64	65	68	69	70	71	72	72	75	75	80	81	83	85
yes	no	yes	yes	yes	no	no	yes	yes	yes	no	yes	yes	no

Discretising temperature in the weather data

- Step 1: order attribute values and find accompanying class
- Step 2: Find breaking points in this sequence:
  - Per class change
  - Impose a minimum number of cases for the majority class
    - O(3)
    - When adjacent partitions have the same majority class they can be merged.

64	65	68	69	70	71	72	72	75	75	80	81	83	85
yes	no	yes	yes	yes	no	no	yes	yes	yes	no	yes	yes	no

Discretising temperature in the weather data

- Step 1: order attribute values and find accompanying class
- Step 2: Find breaking points in this sequence:
  - Per class change
  - Impose a minimum number of cases for the majority class
    - O(3)
    - When adjacent partitions have the same majority class they can be merged.

64	65	68	69	70	71	72	72	75	75	80	81	83	85
yes	no	yes	yes	yes	no	no	yes	yes	yes	no	yes	yes	no

Discretising temperature in the weather data

- Step 1: order attribute values and find accompanying class
- Step 2: Find breaking points in this sequence:
  - Per class change (good idea?)
  - Impose a minimum number of cases for the majority class
    - O(3)
    - When adjacent partitions have the same majority class they can be merged.

64	65	68	69	70	71	72	72	75	75	80	81	83	85
yes	no	yes	yes	yes	no	no	yes	yes	yes	no	yes	yes	no

Discretising temperature in the weather data

- Step 1: order attribute values and find accompanying class
- Step 2: Find breaking points in this sequence:
  - Per class change (good idea?)
  - Impose a minimum number of cases for the majority class
    - O(3)
    - When adjacent partitions have the same majority class they can be merged.

64	65	68	69	70	71	72	72	75	75	80	81	83	85
yes	no	yes	yes	yes	no	no	yes	yes	yes	no	yes	yes	no

1R

- 1R only looks at 1 attribute

- Lets take all attributes into account
  - Naïve bayes

Let us use all the attributes

- Assumptions
  - Equal importance
  - Independence
- Main idea
  - Use the training data for estimating the conditional probabilities:

P(play=YES | (outlook=A, temp=B, humidity=C, wind=D))

P(play=NO | (outlook=A, temp=B, humidity=C, wind=D))

- Use ***Bayes*** formula for this

  - Simplify the estimation process by making a ***naive*** assumption that attributes are independent

Bayes rule

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

Likelihood of play = YES VS likelihood of play = NO

Bayes rule

$$\underline{P(A|B)} = \frac{P(B|A)P(A)}{P(B)}$$

P(play=YES | (sunny, cool, high, true)) =

P((sunny, cool, high, true) | play = YES) *

Bayes rule

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

P(play=YES | (sunny, cool, high, true)) =

P((sunny, cool, high, true) | play = YES) * P(play = YES)

Bayes rule

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

P(play=YES | (sunny, cool, high, true)) =

$$\frac{\textbf{P((sunny, cool, high, true) | play = YES) * P(play = YES)}}{\textbf{P(sunny, cool, high, true)}}$$

Bayes rule

P(play=YES | (sunny, cool, high, true)) =

$$\frac{P(\text{ (sunny, cool, high, true) } \mid \text{ play = YES}) * P(\text{play = YES})}{P(\text{sunny, cool, high, true})}$$

Independence

P(sunny | YES) * P(cool | YES)

Bayes rule

$$P(\text{play=YES} \mid (\text{sunny, cool, high, true})) =$$

$$\frac{P((\text{sunny, cool, high, true}) \mid \text{play = YES}) * P(\text{play = YES})}{P(\text{sunny, cool, high, true})}$$

Independence

$$\frac{P(\text{sunny} \mid \text{YES}) * P(\text{cool} \mid \text{YES}) * P(\text{high} \mid \text{YES}) * P(\text{true} \mid \text{YES}) * P(\text{play = YES})}{P(\text{sunny}) * P(\text{cool}) * P(\text{high}) * P(\text{true})}$$

Bayes rule

P(play=YES | (sunny, cool, high, true)) =

$$\frac{P(\text{(sunny, cool, high, true)} \mid \text{play} = \text{YES}) * P(\text{play} = \text{YES})}{P(\text{sunny, cool, high, true})}$$

Independence

$$\frac{P(\text{sunny} \mid \text{YES}) * P(\text{cool} \mid \text{YES}) * P(\text{high} \mid \text{YES}) * P(\text{true} \mid \text{YES}) * P(\text{play} = \text{YES})}{P(\text{sunny}) * P(\text{cool}) * P(\text{high}) * P(\text{true})}$$

Next step: fill in the numbers

Bayes rule

Table 1.2 **The weather data.**

outlook	temperature	humidity	windy	play
sunny	hot	high	false	no
sunny	hot	high	true	no
overcast	hot	high	false	yes
rainy	mild	high	false	yes
rainy	cool	normal	false	yes
rainy	cool	normal	true	no
overcast	cool	normal	true	yes
sunny	mild	high	false	no
sunny	cool	normal	false	yes
rainy	mild	normal	false	yes
sunny	mild	normal	true	yes
overcast	mild	high	true	yes
overcast	hot	normal	false	yes
rainy	mild	high	true	no

Table 4.2 **The weather data, with counts and probabilities.**

outlook			temperature			humidity			windy			play	
	yes	*no*		*yes*	*no*		*yes*	*no*		*yes*	*no*	*yes*	*no*
sunny	2	3	hot	2	2	high	3	4	false	6	2	9	5
overcast	4	0	mild	4	2	normal	6	1	true	3	3		
rainy	3	2	cool	3	1								
sunny	2/9	3/5	hot	2/9	2/5	high	3/9	4/5	false	6/9	2/5	9/14	5/14
overcast	4/9	0/5	mild	4/9	2/5	normal	6/9	1/5	true	3/9	3/5		
rainy	3/9	2/5	cool	3/9	1/5								

outlook	temperature	humidity	windy	play
sunny	cool	high	true	?

Bayes rule

Table 1.2 The weather data.

outlook	temperature	humidity	windy	play
sunny	hot	high	false	no
sunny	hot	high	true	no
overcast	hot	high	false	yes
rainy	mild	high	false	yes
rainy	cool	normal	false	yes
rainy	cool	normal	true	no
overcast	cool	normal	true	yes
sunny	mild	high	false	no
sunny	cool	normal	false	yes
rainy	mild	normal	false	yes
sunny	mild	normal	true	yes
overcast	mild	high	true	yes
overcast	hot	normal	false	yes
rainy	mild	high	true	no

Table 4.2 The weather data, with counts and probabilities.

outlook			temperature			humidity			windy			play	
	yes	*no*		*yes*	*no*		*yes*	*no*		*yes*	*no*	*yes*	*no*
sunny	2	3	hot	2	2	high	3	4	false	6	2	9	5
overcast	4	0	mild	4	2	normal	6	1	true	3	3		
rainy	3	2	cool	3	1								
sunny	2/9	3/5	hot	2/9	2/5	high	3/9	4/5	false	6/9	2/5	9/14	5/14
overcast	4/9	0/5	mild	4/9	2/5	normal	6/9	1/5	true	3/9	3/5		
rainy	3/9	2/5	cool	3/9	1/5								

outlook	temperature	humidity	windy	play
sunny	cool	high	true	?

Table 1.2 The weather data.

outlook	temperature	humidity	windy	play
sunny	hot	high	false	no
sunny	hot	high	true	no
overcast	hot	high	false	yes
rainy	mild	high	false	yes
rainy	cool	normal	false	yes
rainy	cool	normal	true	no
overcast	cool	normal	true	yes
sunny	mild	high	false	no
sunny	cool	normal	false	yes
rainy	mild	normal	false	yes
sunny	mild	normal	true	yes
overcast	mild	high	true	yes
overcast	hot	normal	false	yes
rainy	mild	high	true	no

Bayes rule

Table 4.2 The weather data, with counts and probabilities.

outlook			temperature			humidity			windy			play	
	yes	*no*		*yes*	*no*		*yes*	*no*		*yes*	*no*	*yes*	*no*
sunny	2	3	hot	2	2	high	3	4	false	6	2	9	5
overcast	4	0	mild	4	2	normal	6	1	true	3	3		
rainy	3	2	cool	3	1								
sunny	2/9	3/5	hot	2/9	2/5	high	3/9	4/5	false	6/9	2/5	9/14	5/14
overcast	4/9	0/5	mild	4/9	2/5	normal	6/9	1/5	true	3/9	3/5		
rainy	3/9	2/5	cool	3/9	1/5								

outlook	temperature	humidity	windy	play
sunny	cool	high	true	?

$$P(YES|....) = \frac{P(sunny \mid YES) * P(cool \mid YES) * P(high \mid YES) * P(true \mid YES) * P(play = YES)}{P(sunny) * P(cool) * P(high) * P(true)}$$

$$P(NO|....) = \frac{P(sunny \mid NO) * P(cool \mid NO) * P(high \mid NO) * P(true \mid NO) * P(play = NO)}{P(sunny) * P(cool) * P(high) * P(true)}$$

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Bayes rule

Table 1.2 **The weather data.**

outlook	temperature	humidity	windy	play
sunny	hot	high	false	no
sunny	hot	high	true	no
overcast	hot	high	false	yes
rainy	mild	high	false	yes
rainy	cool	normal	false	yes
rainy	cool	normal	true	no
overcast	cool	normal	true	yes
sunny	mild	high	false	no
sunny	cool	normal	false	yes
rainy	mild	normal	false	yes
sunny	mild	normal	true	yes
overcast	mild	high	true	yes
overcast	hot	normal	false	yes
rainy	mild	high	true	no

Table 4.2 **The weather data, with counts and probabilities.**

outlook			temperature			humidity			windy			play	
	yes	*no*		*yes*	*no*		*yes*	*no*		*yes*	*no*	*yes*	*no*
sunny	2	3	hot	2	2	high	3	4	false	6	2	9	5
overcast	4	0	mild	4	2	normal	6	1	true	3	3		
rainy	3	2	cool	3	1								
sunny	2/9	3/5	hot	2/9	2/5	high	3/9	4/5	false	6/9	2/5	9/14	5/14
overcast	4/9	0/5	mild	4/9	2/5	normal	6/9	1/5	true	3/9	3/5		
rainy	3/9	2/5	cool	3/9	1/5								

outlook	temperature	humidity	windy	play
sunny	cool	high	true	?

Likelihood of yes = **P(sunny | YES) * P(cool | YES) *P(high | YES) *P(true | YES) * P(play = YES)**

Likelihood of no = **P(sunny | NO) * P(cool | NO) *P(high | NO) *P(true | NO) * P(play = NO)**

Bayes rule

Table 1.2 The weather data.

outlook	temperature	humidity	windy	play
sunny	hot	high	false	no
sunny	hot	high	true	no
overcast	hot	high	false	yes
rainy	mild	high	false	yes
rainy	cool	normal	false	yes
rainy	cool	normal	true	no
overcast	cool	normal	true	yes
sunny	mild	high	false	no
sunny	cool	normal	false	yes
rainy	mild	normal	false	yes
sunny	mild	normal	true	yes
overcast	mild	high	true	yes
overcast	hot	normal	false	yes
rainy	mild	high	true	no

Table 4.2 The weather data, with counts and probabilities.

outlook			temperature			humidity			windy			play	
	yes	*no*		*yes*	*no*		*yes*	*no*		*yes*	*no*	*yes*	*no*
sunny	2	3	hot	2	2	high	3	4	false	6	2	9	5
overcast	4	0	mild	4	2	normal	6	1	true	3	3		
rainy	3	2	cool	3	1								
sunny	2/9	3/5	hot	2/9	2/5	high	3/9	4/5	false	6/9	2/5	9/14	5/14
overcast	4/9	0/5	mild	4/9	2/5	normal	6/9	1/5	true	3/9	3/5		
rainy	3/9	2/5	cool	3/9	1/5								

outlook	temperature	humidity	windy	play
sunny	cool	high	true	?

Likelihood of yes | {sunny,cool,high,true} = 2/9 · 3/9 · 3/9 · 3/9 · 9/14 = 0.0053

Likelihood of no | {sunny,cool,high,true} = ______________________ = 0.0206

Table 1.2 The weather data.

outlook	temperature	humidity	windy	play
sunny	hot	high	false	no
sunny	hot	high	true	no
overcast	hot	high	false	yes
rainy	mild	high	false	yes
rainy	cool	normal	false	yes
rainy	cool	normal	true	no
overcast	cool	normal	true	yes
sunny	mild	high	false	no
sunny	cool	normal	false	yes
rainy	mild	normal	false	yes
sunny	mild	normal	true	yes
overcast	mild	high	true	yes
overcast	hot	normal	false	yes
rainy	mild	high	true	no

Bayes rule

Table 4.2 The weather data, with counts and probabilities.

outlook			temperature			humidity			windy			play	
	yes	*no*		*yes*	*no*		*yes*	*no*		*yes*	*no*	*yes*	*no*
sunny	2	3	hot	2	2	high	3	4	false	6	2	9	5
overcast	4	0	mild	4	2	normal	6	1	true	3	3		
rainy	3	2	cool	3	1								
sunny	2/9	3/5	hot	2/9	2/5	high	3/9	4/5	false	6/9	2/5	9/14	5/14
overcast	4/9	0/5	mild	4/9	2/5	normal	6/9	1/5	true	3/9	3/5		
rainy	3/9	2/5	cool	3/9	1/5								

outlook	temperature	humidity	windy	play
sunny	cool	high	true	?

Likelihood of yes | {sunny,cool,high,true} = 2/9 · 3/9 · 3/9 · 3/9 · 9/14 = 0.0053

Likelihood of no | {sunny,cool,high,true} = ____________________ = 0.0206

There could be a problem in calculation. Do you see it?

The Laplace estimator

- Problem
  - Given the data if one of $P(A_i|yes) = 0$
  - then $P(yes|A_1,\ldots,A_k) = 0$!

Table 4.4 The numeric weather data with summary statistics.

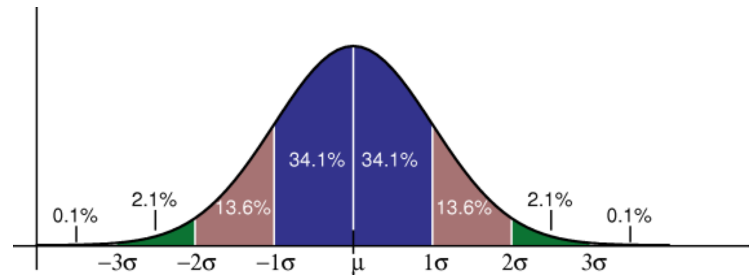
outlook			temperature		humidity		windy			play	
	yes	*no*	*yes*	*no*	*yes*	*no*		*yes*	*no*	*yes*	*no*
sunny	2	3	83	85	86	85	false	6	2	9	5
overcast	4	0	70	80	96	90	true	3	3		
rainy	3	2	68	65	80	70					
			64	72	65	95					
			69	71	70	91					
			75		80						
			75		70						
			72		90						
			81		75						
sunny	2/9	3/5					false	6/9	2/5	9/14	5/14
overcast	4/9	0/5					true	3/9	3/5		
rainy	3/9	2/5									

Table 4.5 Another new day.

outlook	temperature	humidity	windy	play
sunny	66	90	true	?

NB & numerical data

- Solution A:
  - apply any discretization procedure (see slides before)



- Solution B:
  1. assume some distribution (usually normal)
  2. estimate parameters of this distribution (e.g. mean and standard deviation)
  3. use your distribution to estimate conditional probability
  4. plug the probability into the NB formula

$$f(x) = \frac{e^{-(x-\mu)^2/(2\sigma^2)}}{\sigma\sqrt{2\pi}}$$

μ = mean
σ = standard deviation

Table 4.4 **The numeric weather data with summary statistics.**

outlook			temperature			humidity			windy			play	
	yes	*no*		*yes*	*no*		*yes*	*no*		*yes*	*no*	*yes*	*no*
sunny	2	3		83	85		86	85	false	6	2	9	5
overcast	4	0		70	80		96	90	true	3	3		
rainy	3	2		68	65		80	70					
				64	72		65	95					
				69	71		70	91					
				75			80						
				75			70						
				72			90						
				81			75						
sunny	2/9	3/5	*mean*	73	74.6	*mean*	79.1	86.2	false	6/9	2/5	9/14	5/14
overcast	4/9	0/5	*std dev*	6.2	7.9	*std dev*	10.2	9.7	true	3/9	3/5		
rainy	3/9	2/5											

Table 4.5 **Another new day.**

outlook	temperature	humidity	windy	play
sunny	66	90	true	?

Bayes rule

- Not the best performing classifier
- But, good as a benchmark (assignment)

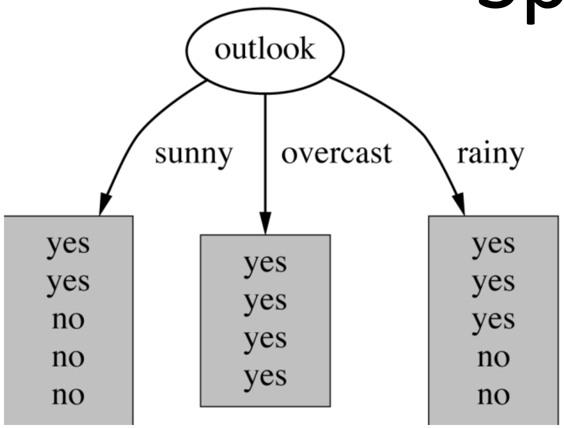
Decision tree

- **Main idea:** Recursively split the data by testing values of attributes (= divide and conquer):
  - **root:** the most *informative* attribute for the whole data set
  - **each child:** the most *informative* attribute for the relevant subset
  - **stop:** when nodes are pure, or when no further split is possible
- How to determine?

Decision tree

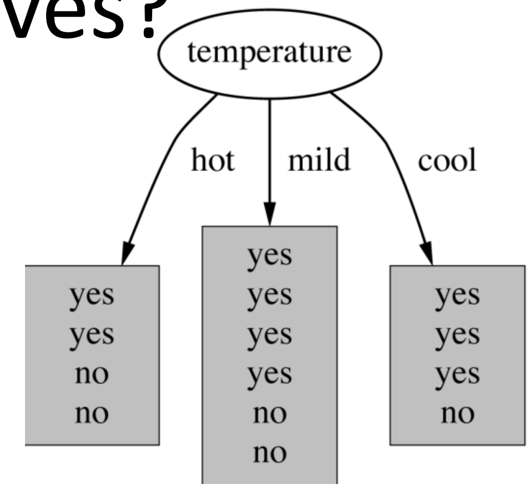
- **Main idea:** Recursively split the data by testing values of attributes (= divide and conquer):
  - **root:** the most *informative* attribute for the whole data set
  - **each child:** the most *informative* attribute for the relevant subset
  - **stop:** when nodes are pure, or when no further split is possible
- Measures of ***informativeness***:
  - information gain

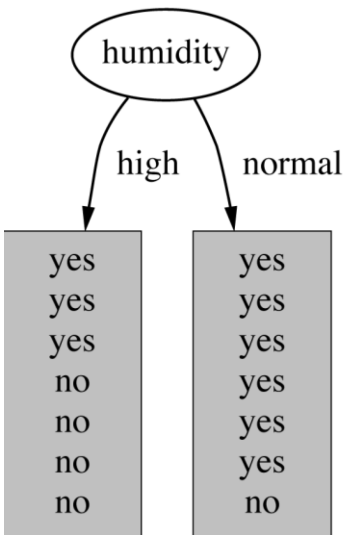
Split? Possible leaves?



A

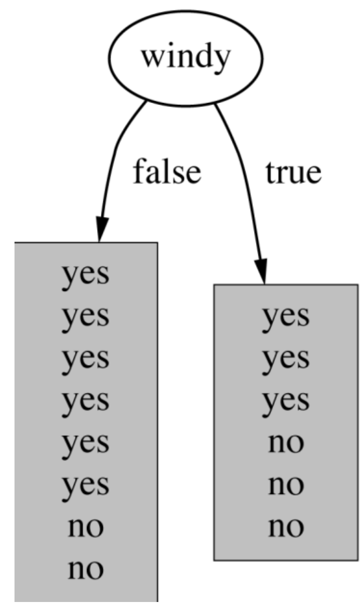
B



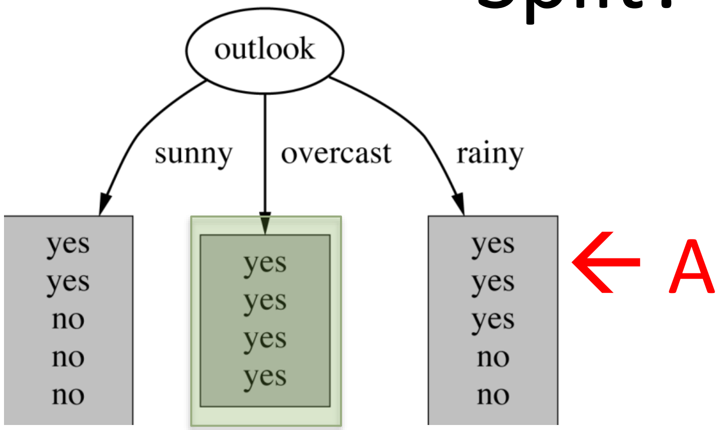


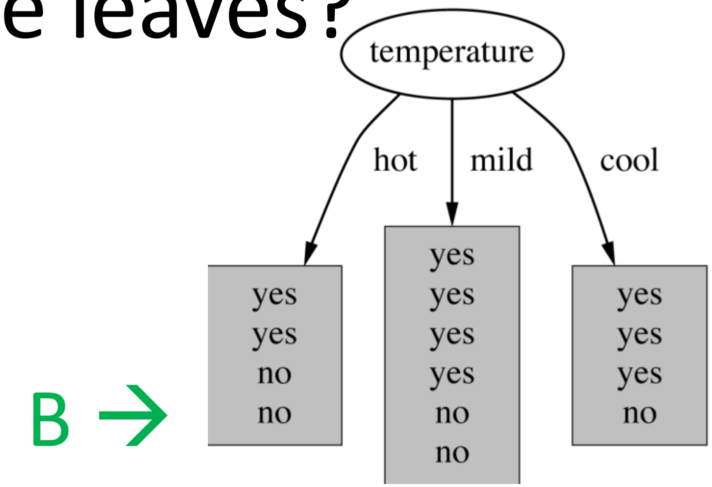
C

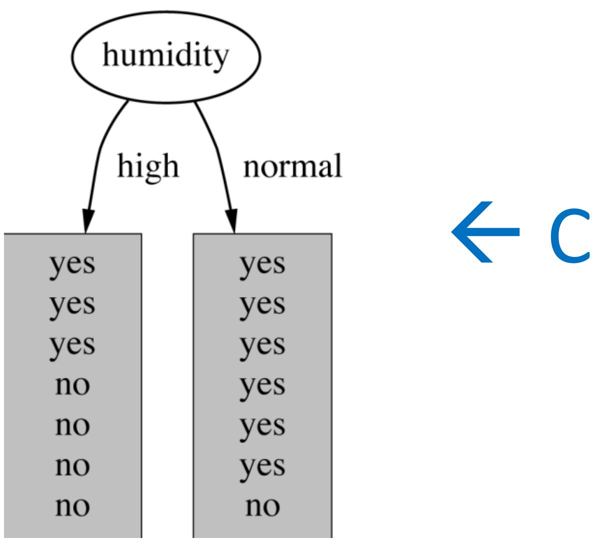
D

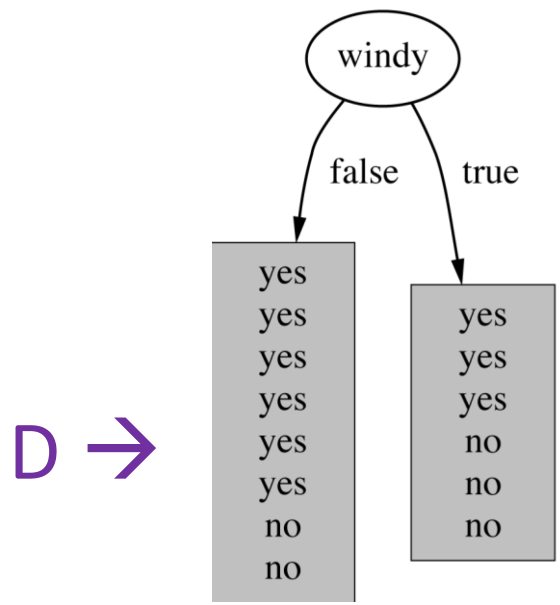


Split? Possible leaves?

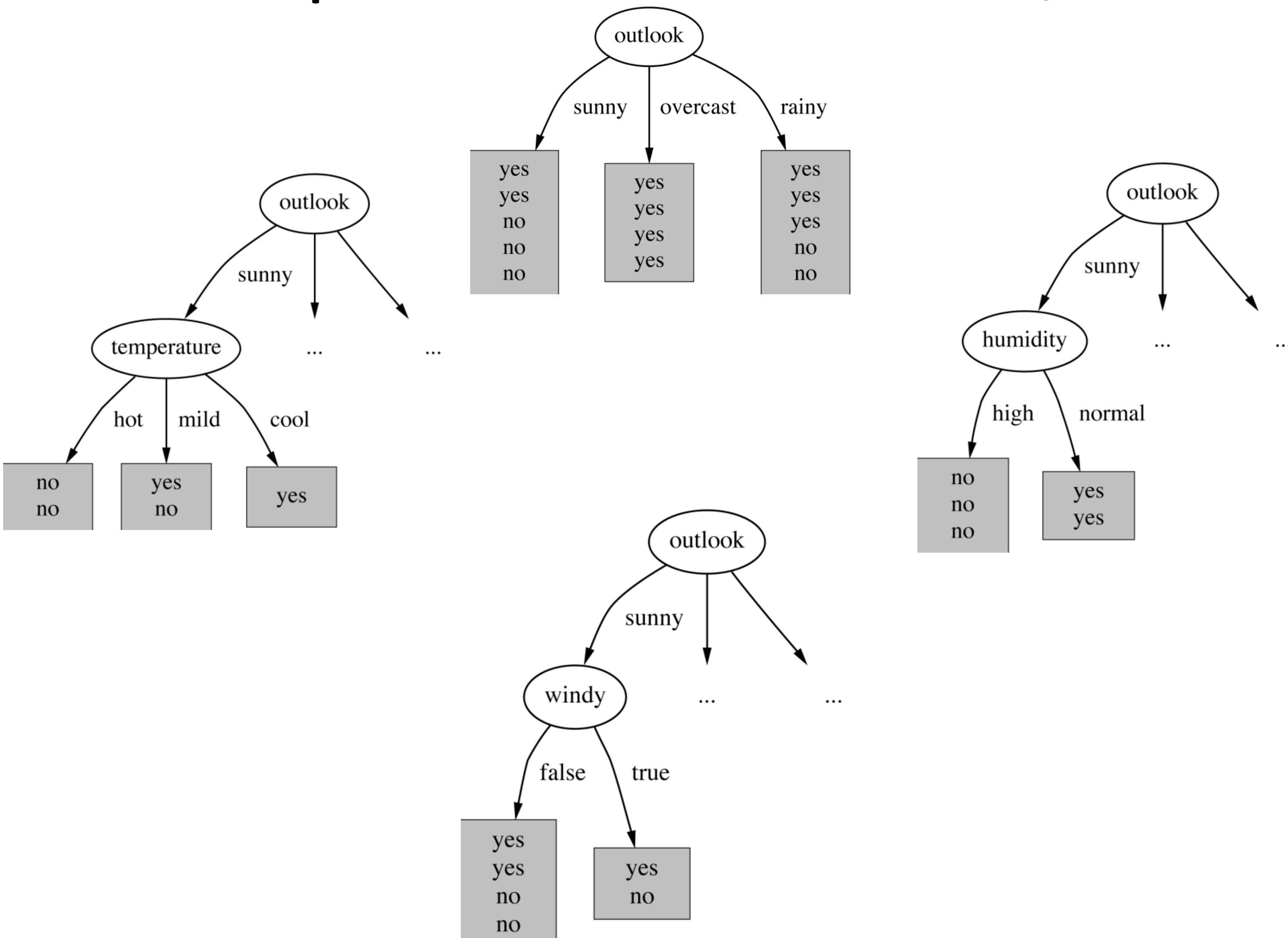




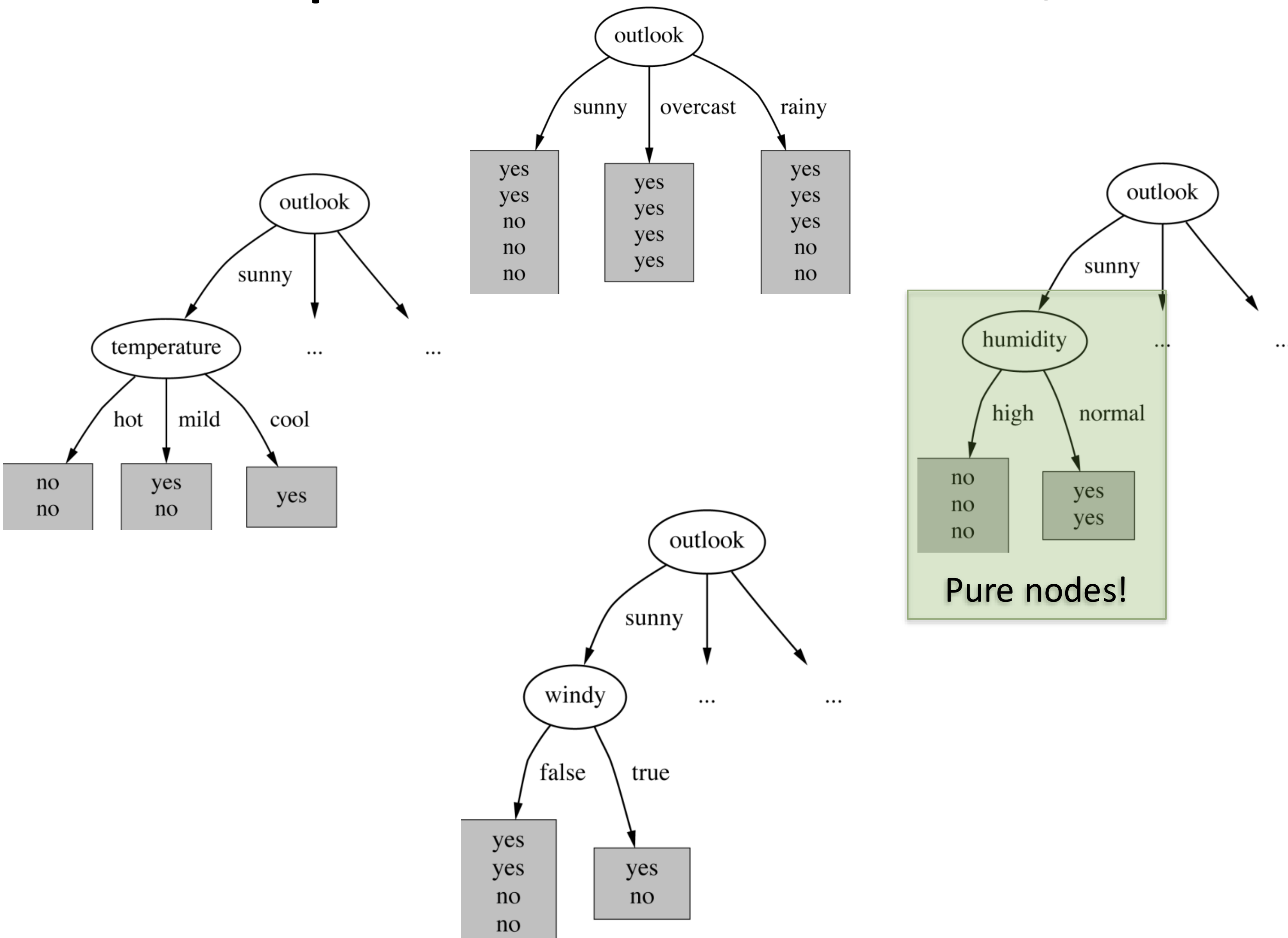




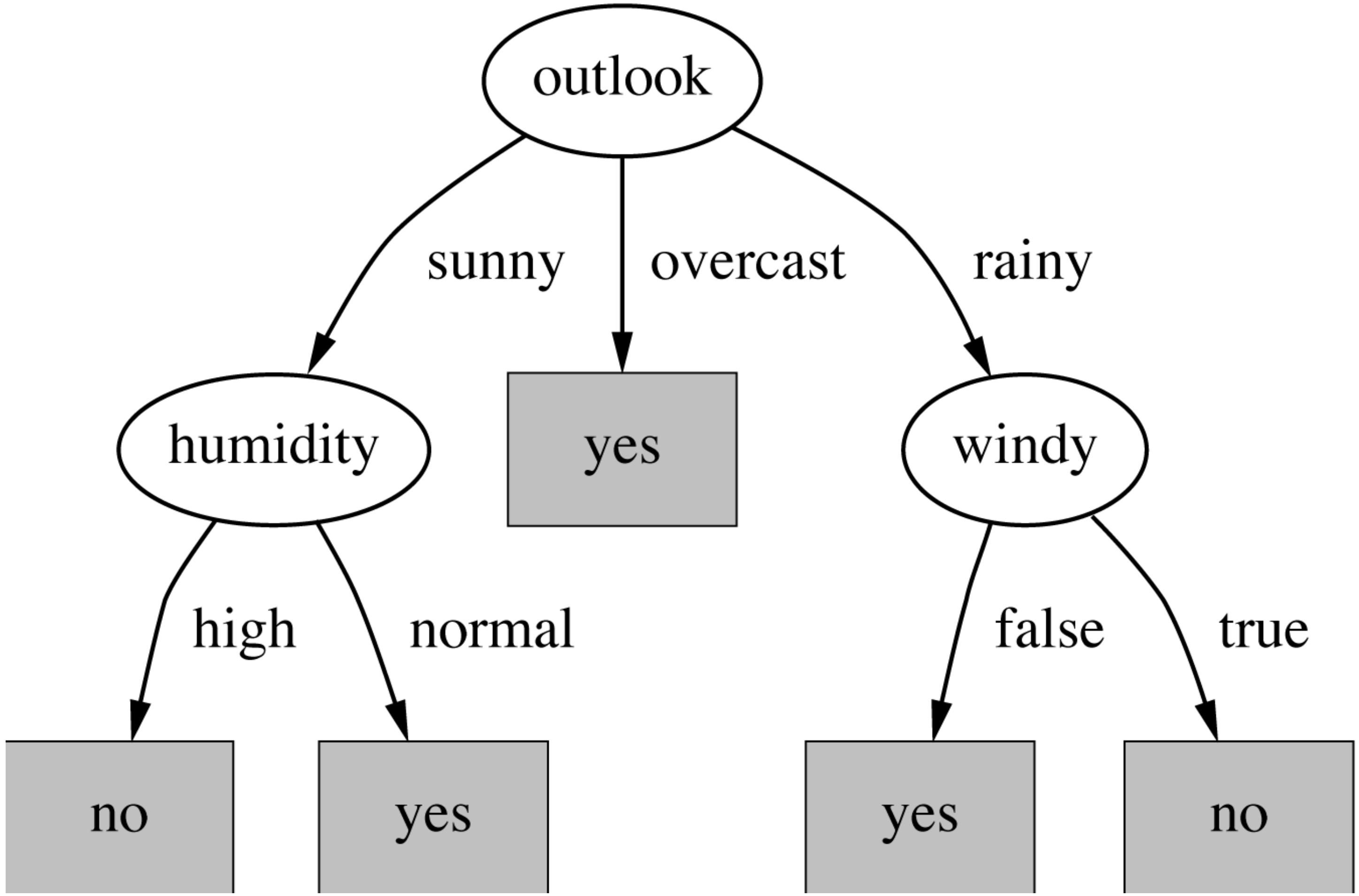
Next split (on the *outlook = sunny* branch)



Next split (on the *outlook = sunny* branch)



The final decision tree



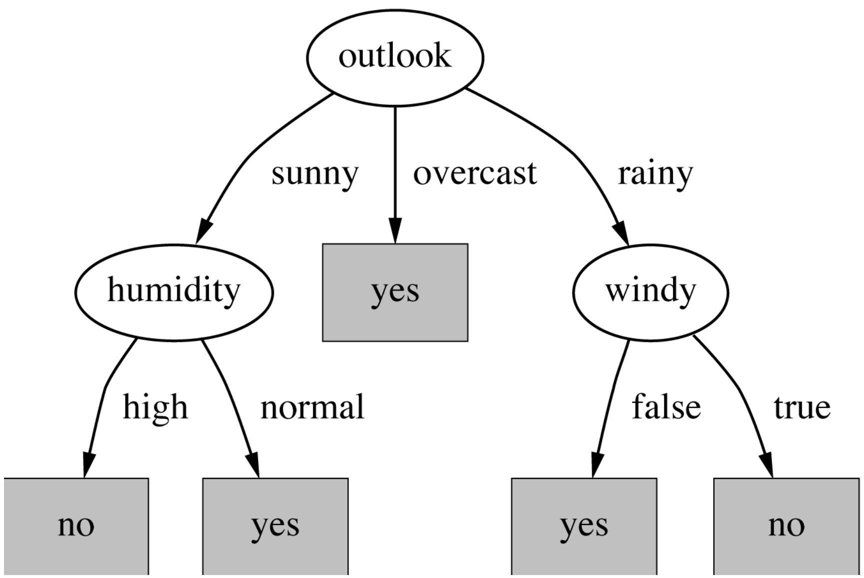
Information gain

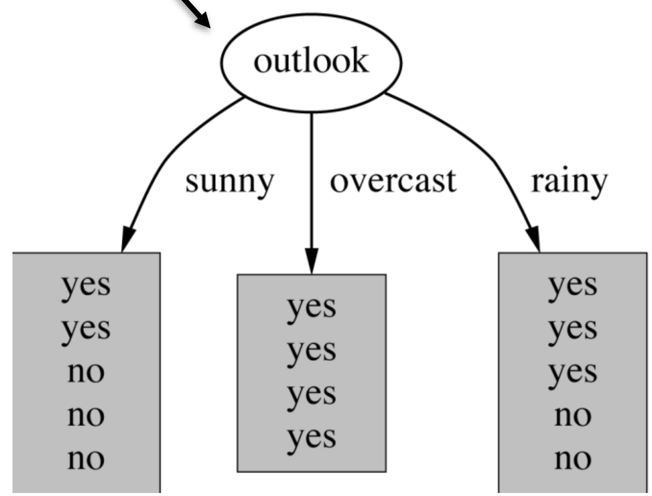
- Weather before splits:

  yes: 9, no: 5

- After a split on outlook:

  sunny (2,3) + overcast (4,0) + rainy (3,2)





Information gain

- Weather before splits:

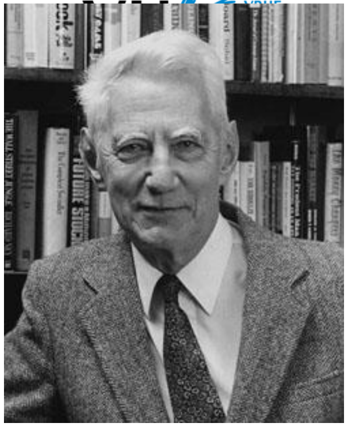
  yes: 9, no: 5

- After a split on outlook:

  sunny (2,3) + overcast (4,0) + rainy (3,2)

- Information gain = info(before) – info(after)
- Info() requirements:
  - info([n,0]) = info([0,n]) = 0
  - info([n,n]) = maximum info value possible
  - info([2,3,4]) = info([2,7]) + 7/9·info([3,4])

Information value



- Aka. *entropy*

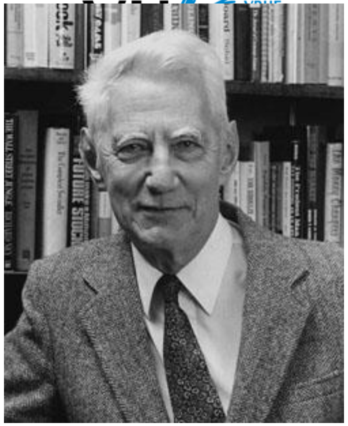
$$\text{entropy}(p_1, p_2, \ldots, p_n) = -p_1 \cdot \log p_1 - p_2 \cdot \log p_2 - \ldots - p_n \cdot \log p_n$$

- Weather before splits:

yes: 9, **no:** 5

  - $\text{Info}([9,5]) = -9/14 * \log(9/14) - 5/14 * \log(5/14)$

    $= 0.4098 + 0.5305 = 0.940 \rightarrow$ *Original*



Information value

- Aka. *entropy*

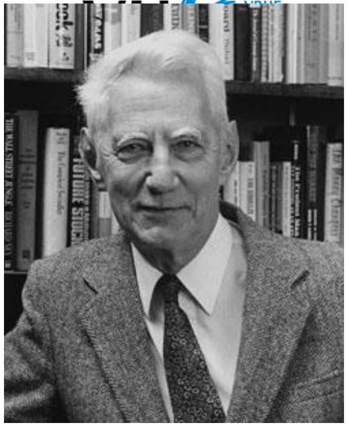
$$\text{entropy}(p_1, p_2,\ldots, p_n) = -p_1 \cdot \log p_1 - p_2 \cdot \log p_2 - \ldots - p_n \cdot \log p_n$$

- Weather data:
  - Info([9,5]) = 0.940 → *Original*
- After a split on outlook:

  sunny (2,3) + overcast (4,0) + rainy (3,2)

  - Info([2,3], [4,0], [3,2]) =

    → *Outlook*



Information value

- Aka. *entropy (log base 2)*

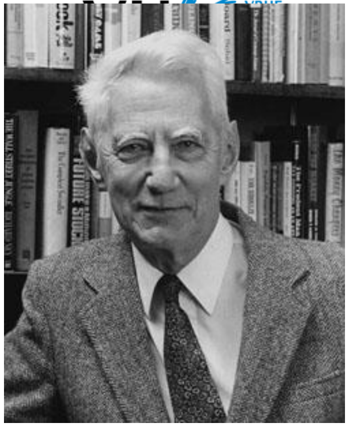
$$\text{entropy}(p_1, p_2,\ldots, p_n) = -p_1 \cdot \log p_1 - p_2 \cdot \log p_2 - \ldots - p_n \cdot \log p_n$$

- Weather data:
  - Info([9,5]) = 0.940 → *Original*
- After a split on outlook:

sunny (2,3) + overcast (4,0) + rainy (3,2)

  - Info([2,3], [4,0], [3,2]) = (5/14)*0.971 + (4/14)*0 + (5/14)*0.971 = 0.693 → *Outlook*

Information value



- Aka. *entropy*

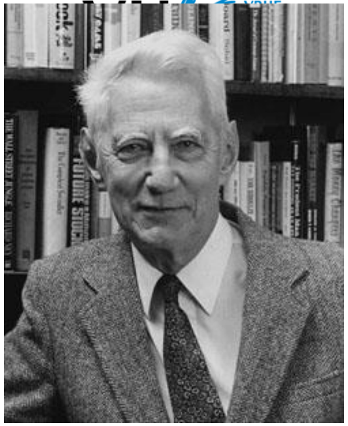
  $$\text{entropy}(p_1, p_2, \ldots, p_n) = -p_1 \cdot \log p_1 - p_2 \cdot \log p_2 - \ldots - p_n \cdot \log p_n$$

- Weather data:
  - Info([9,5]) = 0.940 → *Original*
  - Info([2,3], [4,0], [3,2]) = (5/14)*0.971 + (4/14)*0 + (5/14)*0.971 =

    = 0.693 → *Outlook*

Gain = Original – Outlook =

Information value



- Aka. *entropy*

$$\text{entropy}(p_1, p_2, \ldots, p_n) = -p_1 \cdot \log p_1 - p_2 \cdot \log p_2 - \ldots - p_n \cdot \log p_n$$

- Weather data:
  - Info([9,5]) = 0.940 → *Original*
  - Info([2,3], [4,0], [3,2]) = (5/14)*0.971 + (4/14)*0 + (5/14)*0.971 = = 0.693 → *Outlook*

Gain = Original – Outlook = 0.247, which is higher than for temp. *(0.029)*, humidity *(0.152)* or windy *(0.048)*

What we learned about today...

- Exploratory Data Analysis
- Simple learning algorithms
  - 1Rule algorithm
    - Discretisation
  - Naive Bayes
  - Decision trees
    - Information gain / entropy

Next Lecture...

- More classification algorithms
- Instance-based learning
- Association rules